

Human Computer Interaction

Johan F. HOORN and Jeff TANG

Lecture 05, Part 2: Ergonomics and I/O devices

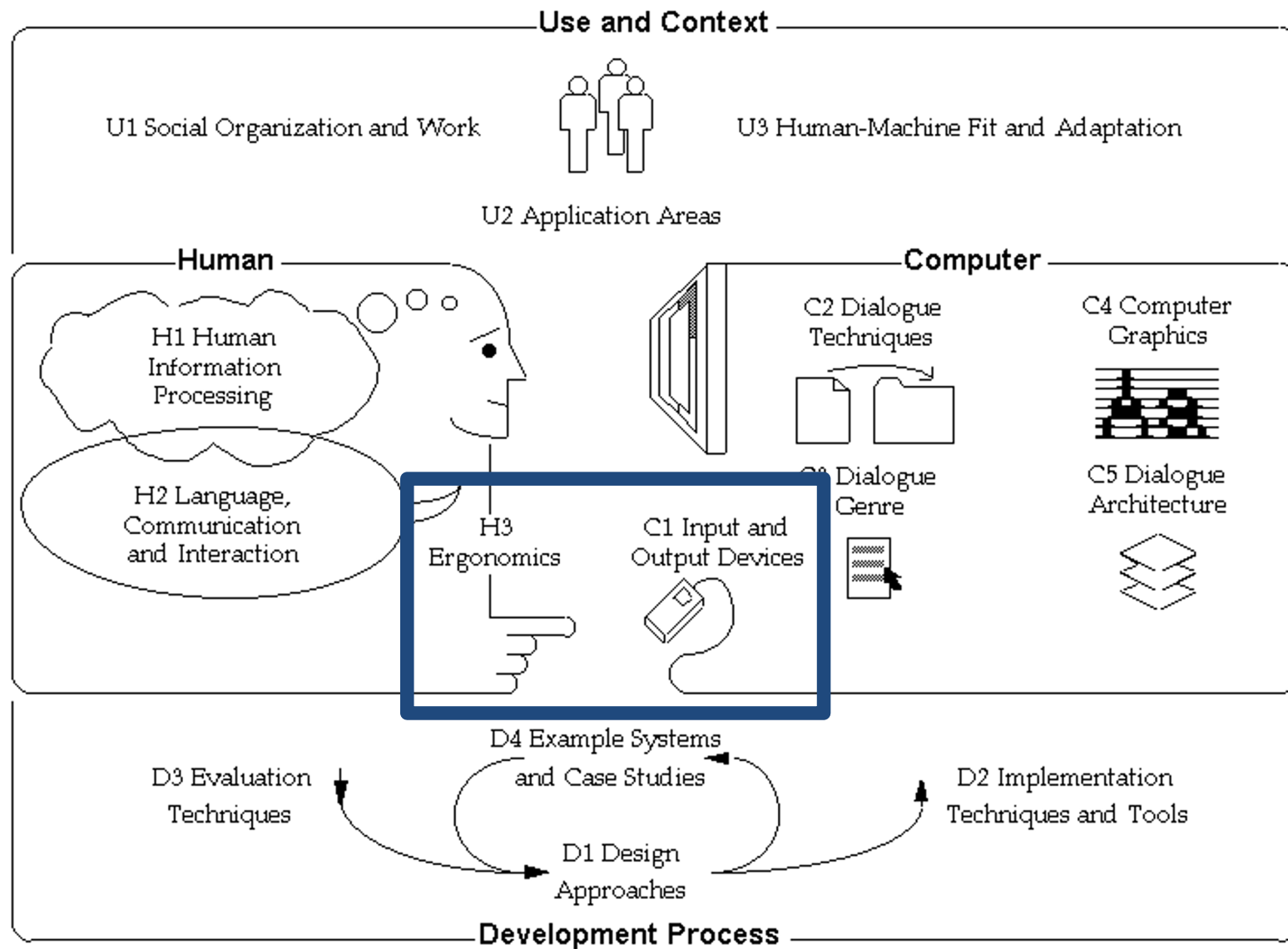


C-3PO ("3PO" aka Threepio)

Lukas, G., & Hales, J. (2002). *Star Wars Episode II: Attack of the Clones*. Los Angeles, CA: 20th Century Fox. [Part 7/10] - 1080p HD, 7:00 min. Retrieved May 21, 2012 from http://www.youtube.com/watch?v=-o5Y_AszZGg



Overview



Contents

- ❑ Ergonomics
- ❑ Input Devices
- ❑ Output Devices
 - Haptics
 - Sound
- ❑ Advanced and futuristic I/O



Slides in this lecture are partially based on those from Ms. Tiffany Tang, Dr. Vincent Ng, Dr. Grace Ngai, Dr. Saul Greenburg, David Ellsworth, Jelle vd Berg, Franz J. Kurfess, and Dr. A. K. Sengupta



I/O devices form the 'user interface.' Let's look at output devices from an ergonomic perspective

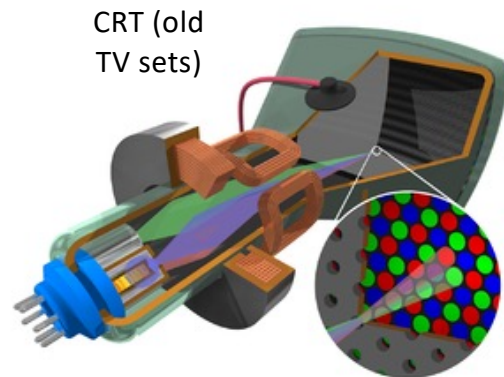
Output



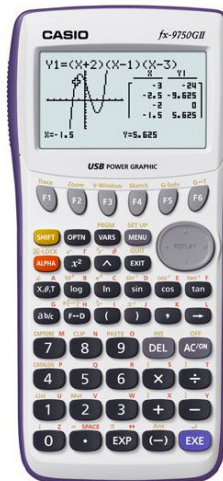
Output devices

☐ Visual

- Character-based displays
 - Liquid-Crystal Displays (LCDs), Flat-Panel Displays (FPDs)
 - Used in stationary devices, in telephones, calculators, etc.
- Graphics displays
 - Cathode Ray Tubes (CRTs), LCDs, and other FPDs, 3D
 - High-resolution graphic displays in stationary output devices (e.g., cockpits, helmet-mounted displays)



Source: Wikipedia



Source: CASIO

This is also
LCD

Windshield serves as
display area



Source: BMW

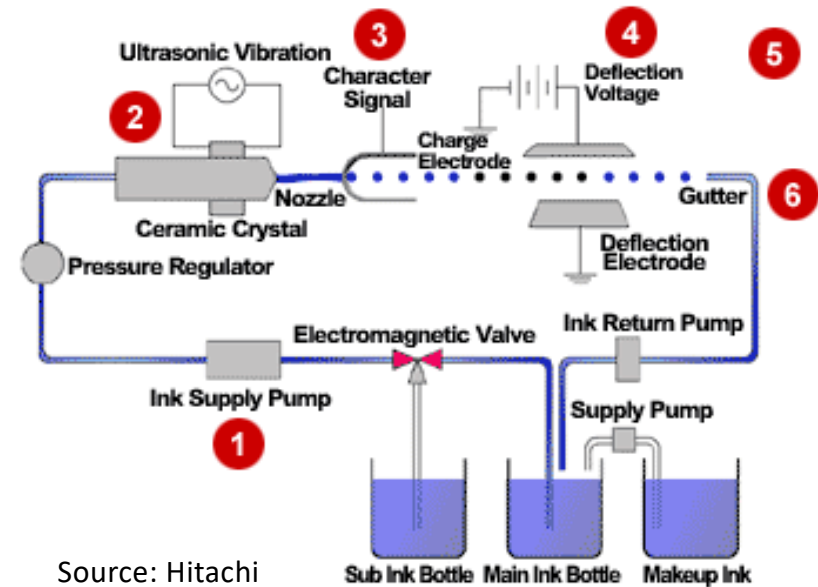
Output devices

Visual

- Printing devices
 - Color vs. b/w, dot matrix, laser jet, inkjet
 - Fax, plotters (colored pens)
 - 3D printing
- Microfiche, micro-film, videotape
 - Require old-school equipment to read

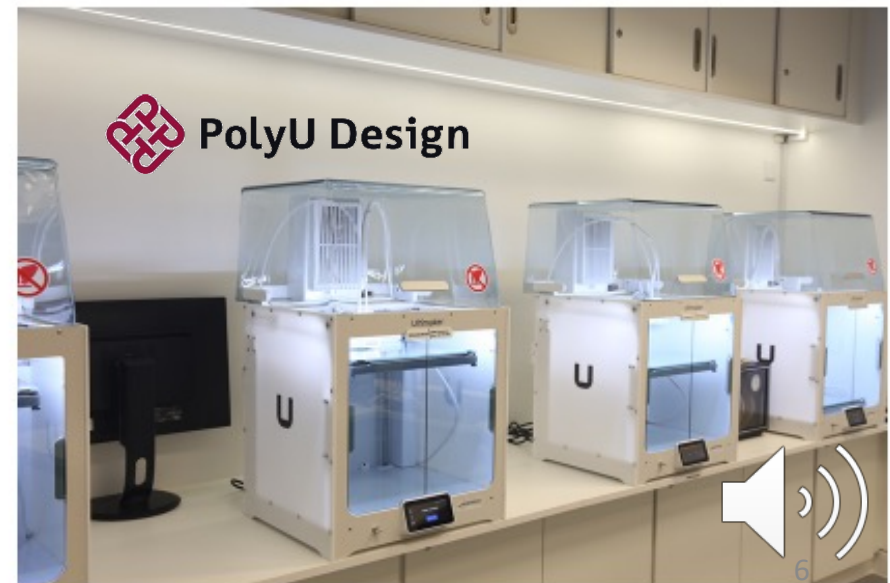


Source: Guana



Source: Hitachi

3D Printing Workshop (1/F)



Output devices

☐ Which portrait was painted by Rembrandt? Left? Right? Both? Neither?



Output devices

☐ Visual

- 3D printing

- Technology

deep learning algorithm

facial recognition technology

- Steps

Step 1: Gathering the paintings from Rembrandt

Step 2: Determining the subject of paintings from Rembrandt

Step 3: Determining the subject of portrait

Step 4: Generating the feature of Rembrandt

Step 5: Identify the facial proportion

Step 6: 3D printing



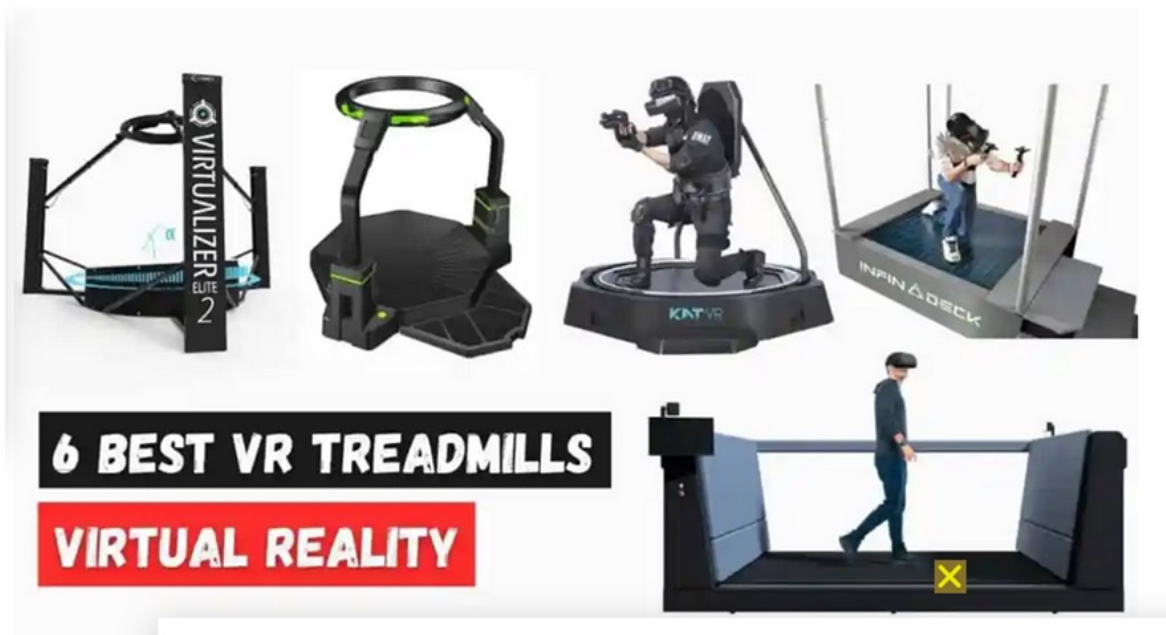
Output devices

❑ Haptic (tactile)

- Force feedback
- Textural feedback (vibro-tactile, electro-tactile, ultrasound)
- Thermal feedback

6 Best VR Treadmills 2021: Omnidirectional Virtual Reality Haptic Treadmill

Virtual Reality Gadgets, Electronic Gadgets, Tech Gadgets, Tech News, VR Games / By techhunterds / March 13, 2021



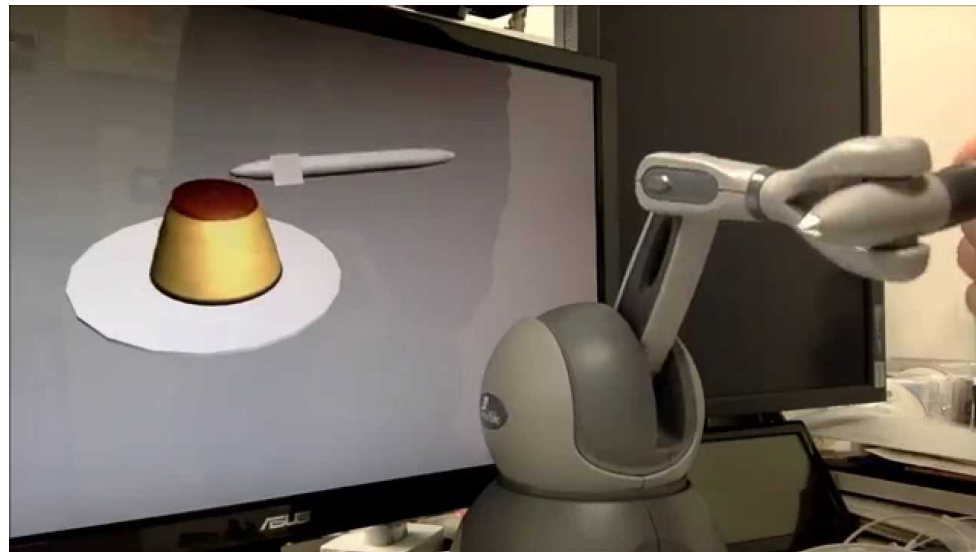
Output devices

□ Haptic (tactile)

- Force feedback is most studied
 - Very realistic in certain scenarios (e.g., in the training of dentists)
 - Commercial products are widely available



Phantom Omni from Sensable Technologies



Here used to design in Unity

Output devices

❑ Haptic (tactile)

- Force feedback is most studied
 - Very realistic in certain scenarios (e.g., virtual tools, orthopedics, surgery)
 - Commercial products are widely available



Force feedback data gloves to practice with virtual tools



Orthopedic simulation

Output devices

❑ Haptic (tactile)

- Textural Feedback
 - Generic feedback of textures is hard to achieve
 - Sensory / perceptual illusions can help create realistic textural feedback
 - Haptic resolution and haptic perception

Here the illusion is created with ultrasound – why is sound also ‘touch?’

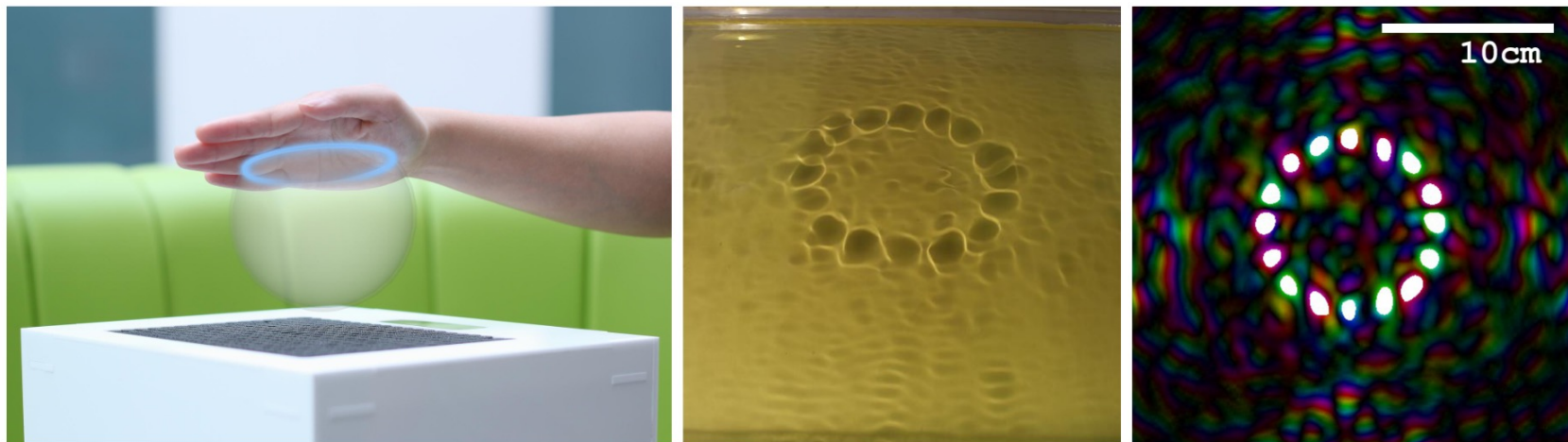


Figure 1: *Left: Ultrasound is focused to create the shape of a virtual sphere. Middle: Ultrasound pushes the impression of a sphere section into oil. Right: A simulation of the ultrasound at the plane intersected by the hand, where a threshold has been applied to highlight the foci.*



Output devices

❑ Haptic (tactile)

- Sensory / perceptual illusions
 - Rubber hand illusion
 - Haptic retargeting (use a single physical object to create the illusion of touching multiple virtual objects)
 - Haptic proxy



Side note: So apart from optical illusions such as depth perception through motion parallax, optic flow, and slanted lines (please check Lecture 03 again), we also anticipate to **feel** something when we **see** it is about to touch us (the brain fills in the associated feeling already)



Tactile illusions (also in VR): the Rubber Hand experiment



Visual support of tactile feedback in data gloves?

Tactile illusions (also in robotics): the Rubber Hand experiment



Exoskeletons? Prostheses (artificial limbs)?

Output devices

□ Haptic (tactile)

- Rubber Hand illusion, what do we learn from it?

- Body transfer illusion:

The illusion of owning either a part of a body or an entire body other than one's own

- Embodiment - three sub-components (Kilteni et al., 2012):

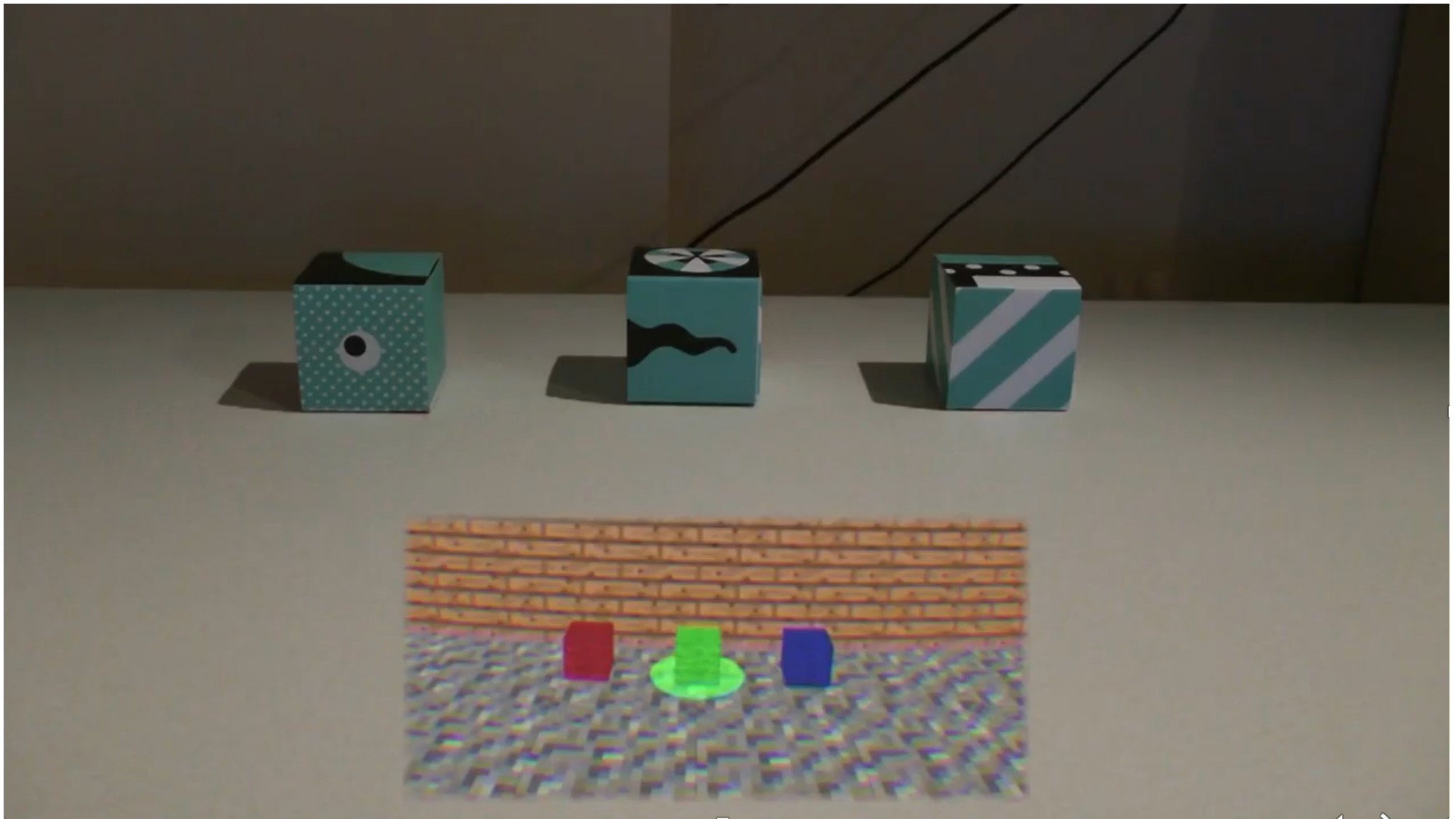
1. Sense of self-location - a determinate volume in space where one feels to be located

2. Sense of agency - the sense of having global motor control, including the subjective experience of action, control, intention, motor selection, and the conscious experience of will

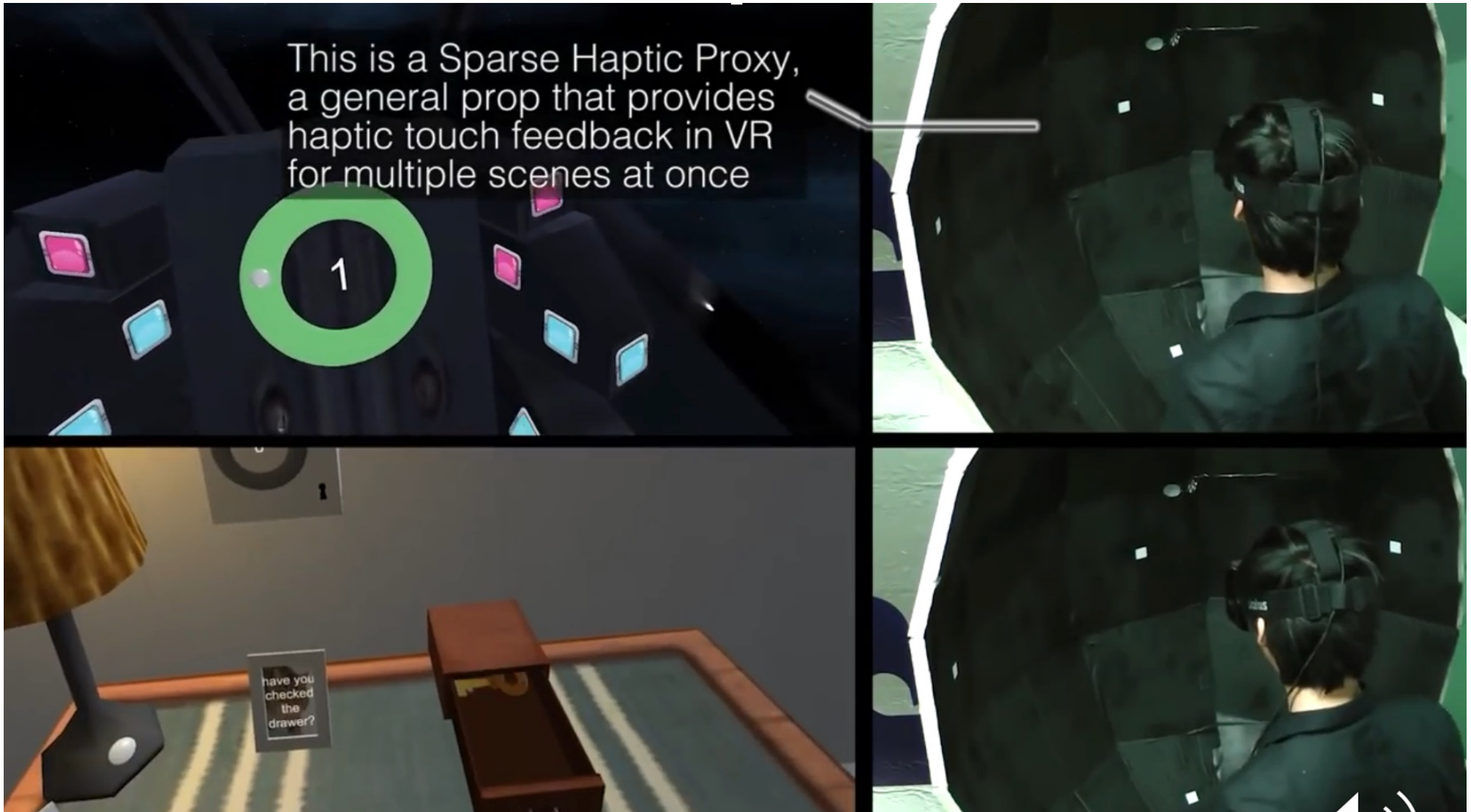
3. Sense of body ownership (your body is 'yours')



Haptic Retargeting



Sparse Haptic Proxy: A saucer that makes the user feel s/he touches buttons projected on a visor



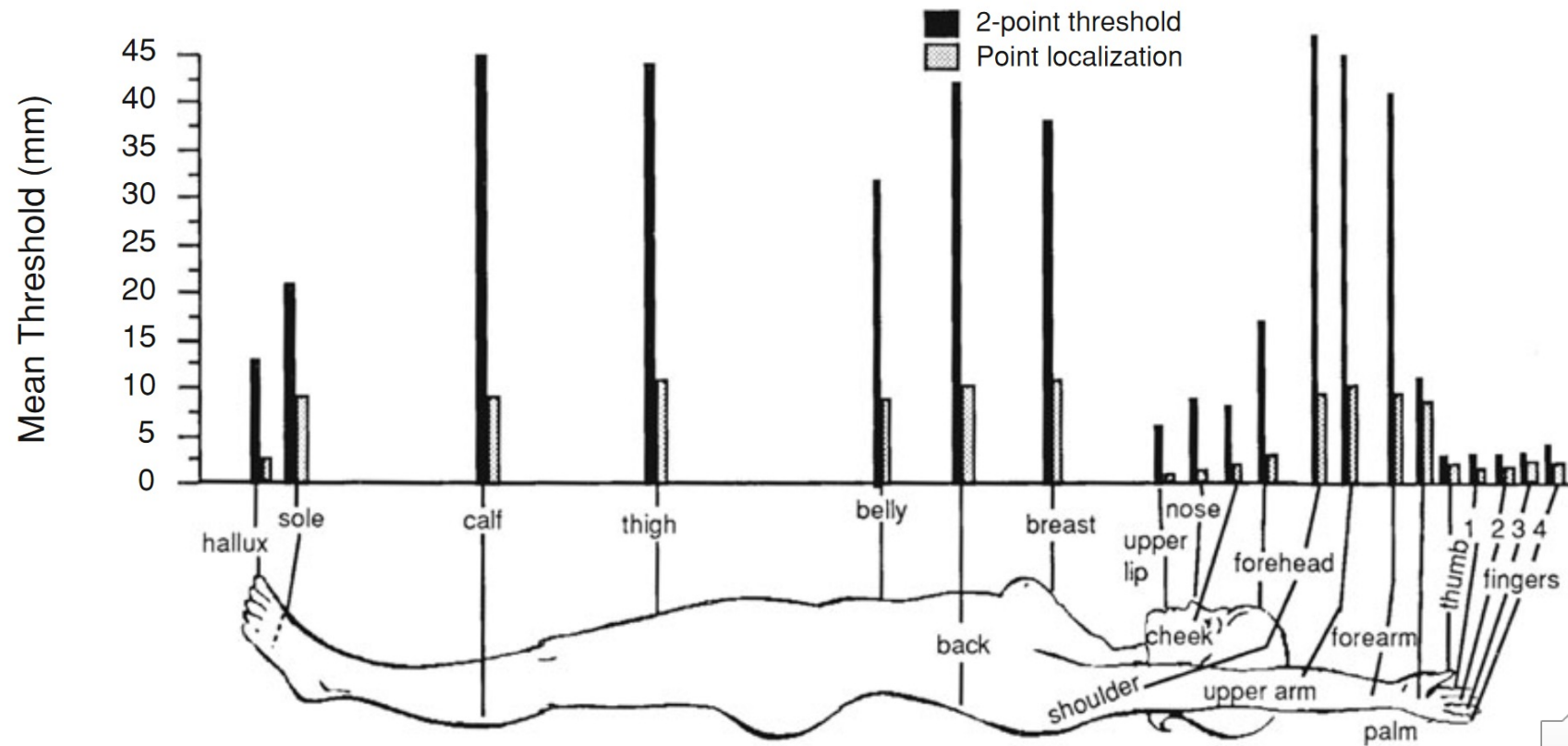
Sparse Haptic Proxy: Fitts' Law active in virtual environments



Output devices

□ Haptic resolution

- Different haptic devices can render haptics in different resolutions
- Different body areas have different spatial and temporal resolving capacity of haptics

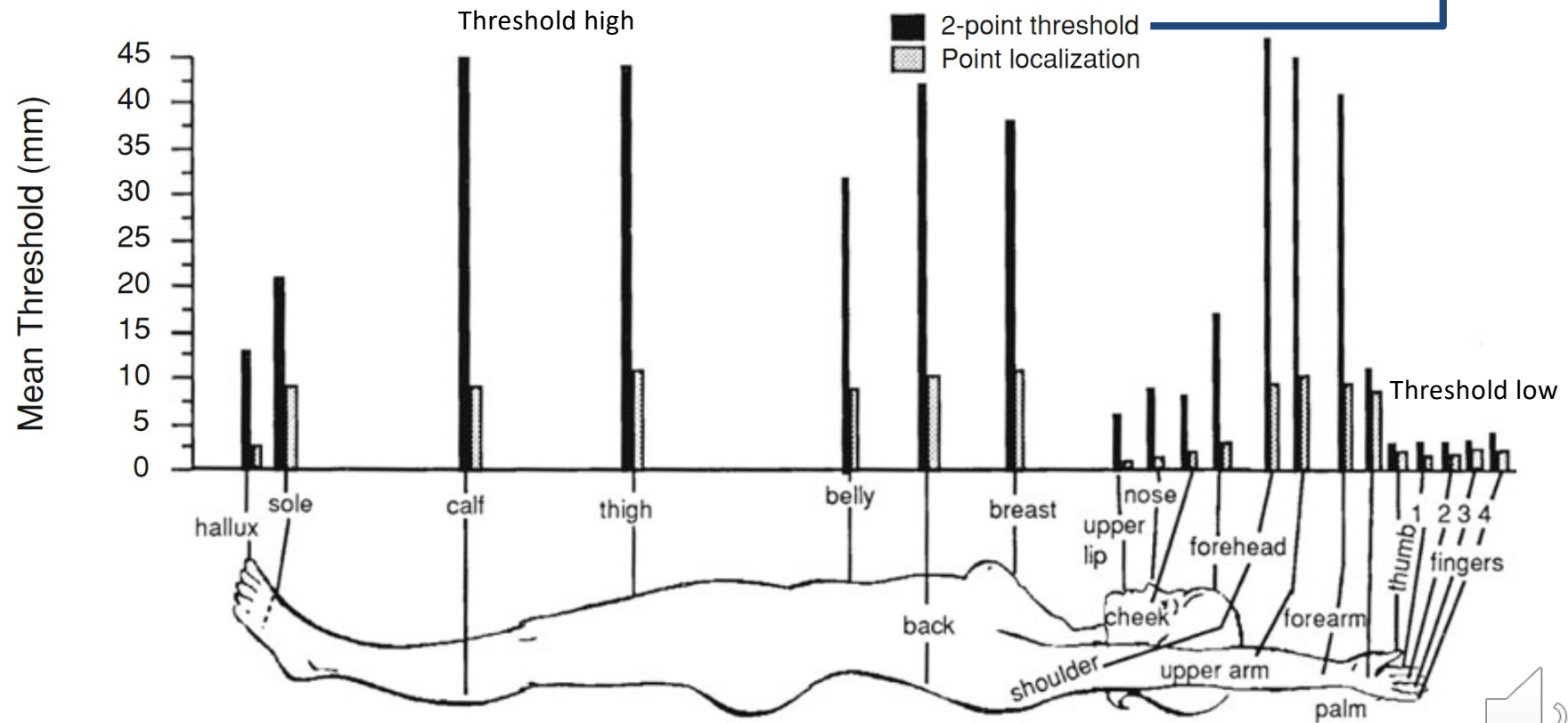


Output devices

□ Haptic resolution

- Different haptic devices can render haptics in different resolutions
- Different body areas have different spatial and temporal resolving capacity of haptics

The two-point touch threshold represents the smallest spatial separation between two stimuli applied to the skin that can be detected some arbitrary percentage of the time.

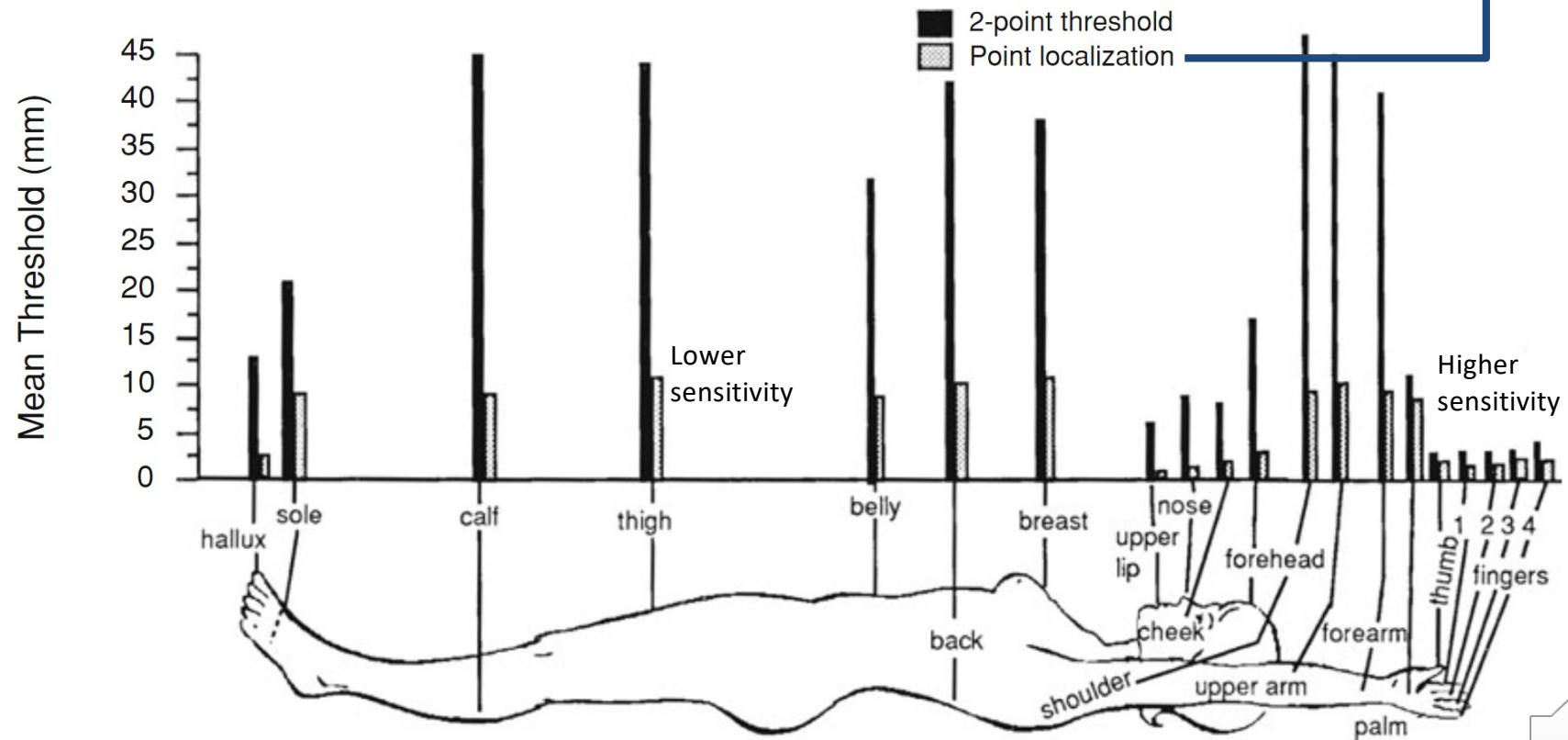


Output devices

□ Haptic resolution

- Different haptic devices can render haptics in different resolutions
- Different body areas have different spatial and temporal resolving capacity of haptics

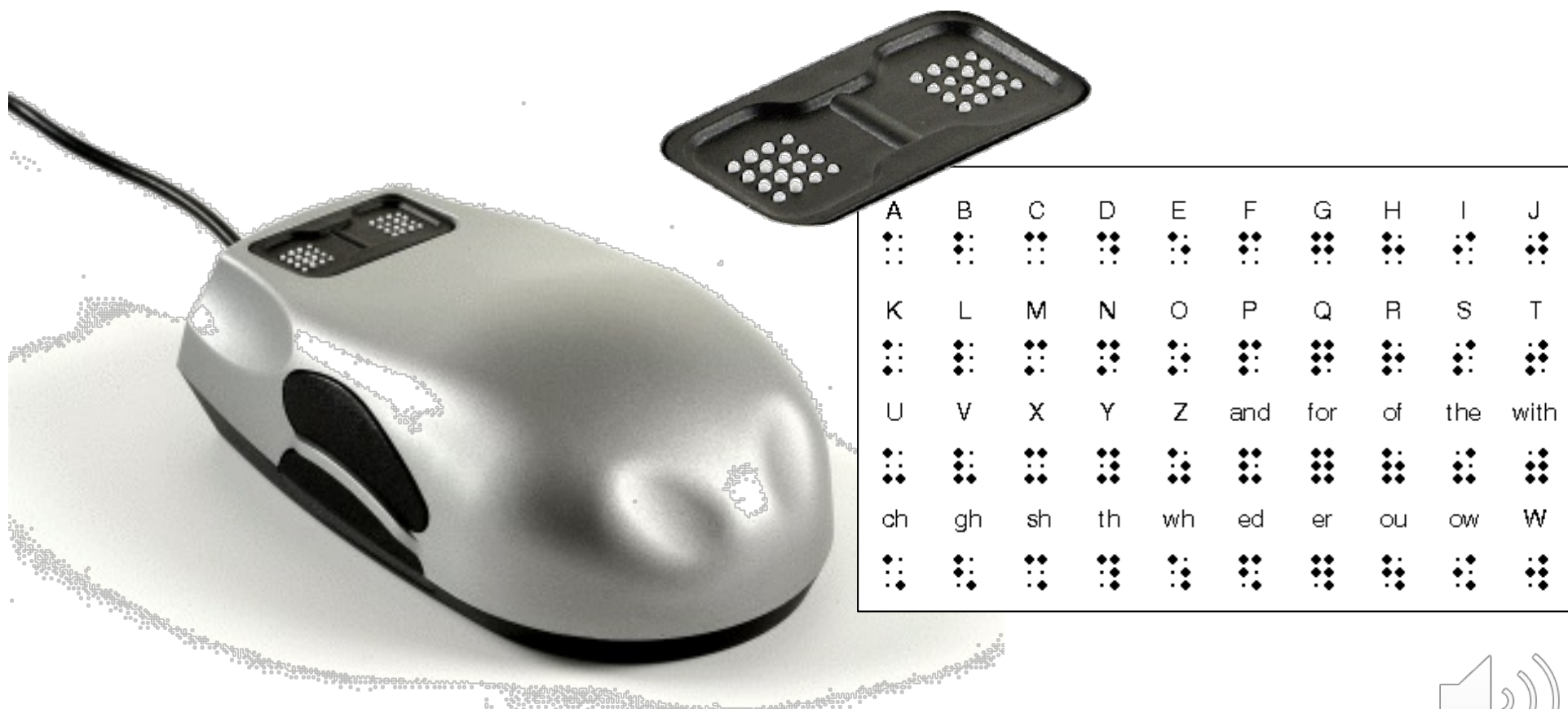
When evaluating the point-localization threshold, a stimulus is presented to the skin, followed in time by a second stimulus that may or may not be applied to the same site. Observers are required to say whether the two stimuli occur at the same or different locations.



Output devices

- ❑ Haptics make for accessible technology

Virtual touch mouse for the visually impaired (mouse as *output* device)



Courtesy Jelle van den Berg

Output devices

❑ Haptics make for accessible technology

The VirTouch mouse has two 4 x 4 pin matrices for tactile feedback. Texts can be read in Braille (one character for each matrix)

The VirTouch mouse also reacts on visual imagery. It uses contrasts (dark colors against light colors) to offer a tactile representation (geographical maps!)

Together, these two functions may be used to scan a display screen for text, layout, but is harder for images



Courtesy Jelle van den Berg

Output devices

- ❑ Haptics make for accessible technology

Tactile exploration is local. Hard to imagine 'the big picture' through the mouse



Better have it 3D printed?

Translates contrasts from a special on-screen map as if it were a tactile map



Courtesy Jelle van den Berg



Output devices

☐ Sound

- Tone generators (beeps, alerting tones)
- Alarms (warning signals)
- Digitized speech (digitally recorded human speech)
- Text-to-speech synthesis
(speech generated by concatenating basic speech sounds according to rules)



Welcome to NaturalReader

Chinese



Yunjian



Xiaoxiao



Yunze



Xiaoqiu



Output devices

□ Sound

- Auditory icons and “earcons”
 - non-speech sounds to inform about current events (e.g., empty trash, new mail, move, copy, delete)
 - should be optional (annoying, may disturb others)
 - corporate earcons (auditory logos), commercial brand-lining, distinctive music (e.g., Macintosh startup sound, Windows launch sound, “Intel Inside”)

Guess the sound:



Clip A



Clip B



Clip C



Clip D



Output devices

□ Auditory interpretation

- Auditory icons and “earcons”

Detection

Is it there?

Attention

What’s up?

Localization

Where from?

Relative discrimination

Are there more?

Identification

It’s this one!

Recognition

Opening tune

E.g., a warning such as a System Error bleep



Output devices

□ Auditory interpretation

- Auditory icons and “earcons”

Detection

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E.g., a warning such as a System Error bleep

E.g., when indicating interaction changes



Output devices

□ Auditory interpretation

- Auditory icons and “earcons”

Detection

Is it there?

Attention

What's up?

Localization

Where from?

Relative discrimination

Are there more?

Identification

It's this one!

Recognition

Opening tune

E.g., a warning such as a System Error bleep

E.g., when indicating interaction changes

E.g., an artificial earcon (melody) is easier distinguished from surrounding sounds than the natural sound of an auditory icon (Sikora et al., 1995)



Output devices

□ Auditory design

- Compatibility: Signals should correspond to natural sound experience
- Approximation: Two-stage signals should bring clear message
 - 1) Basic meaning (e.g., attention)
 - 2) Higher meaning (where, what for, how)
- Parsimony: Signal should provide only necessary information
- Avoid extremes: High intensity signals cause distraction or disruption of response
- Do not overload the auditory system
- Environment: Comply with signal-to-noise ratio
(How noisy is the interface or work environment? Can users hear your sounds?)
- Design trade-off:
 - Invariance: Type of signal should be constant in its meaning (signal used for different purposes is bad → ambiguity!!)
 - Use interrupted or variable signals: Humans adapt to signals and might react less on those signals in the future (cf. Three Mile Island disaster)

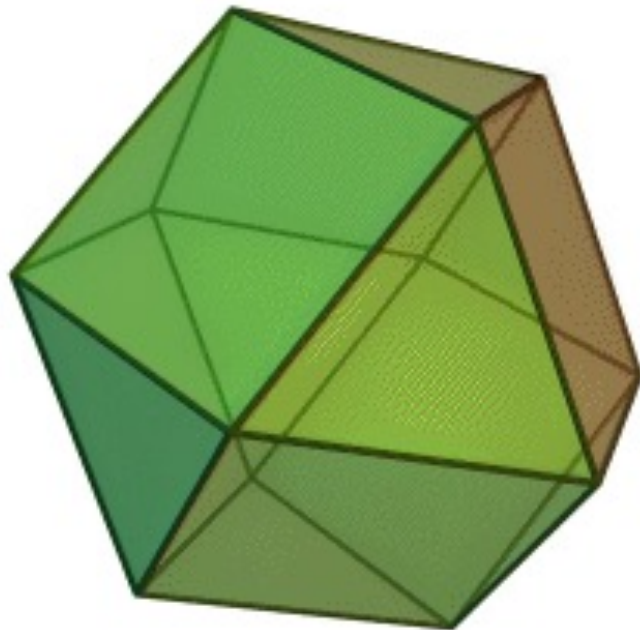


Output devices

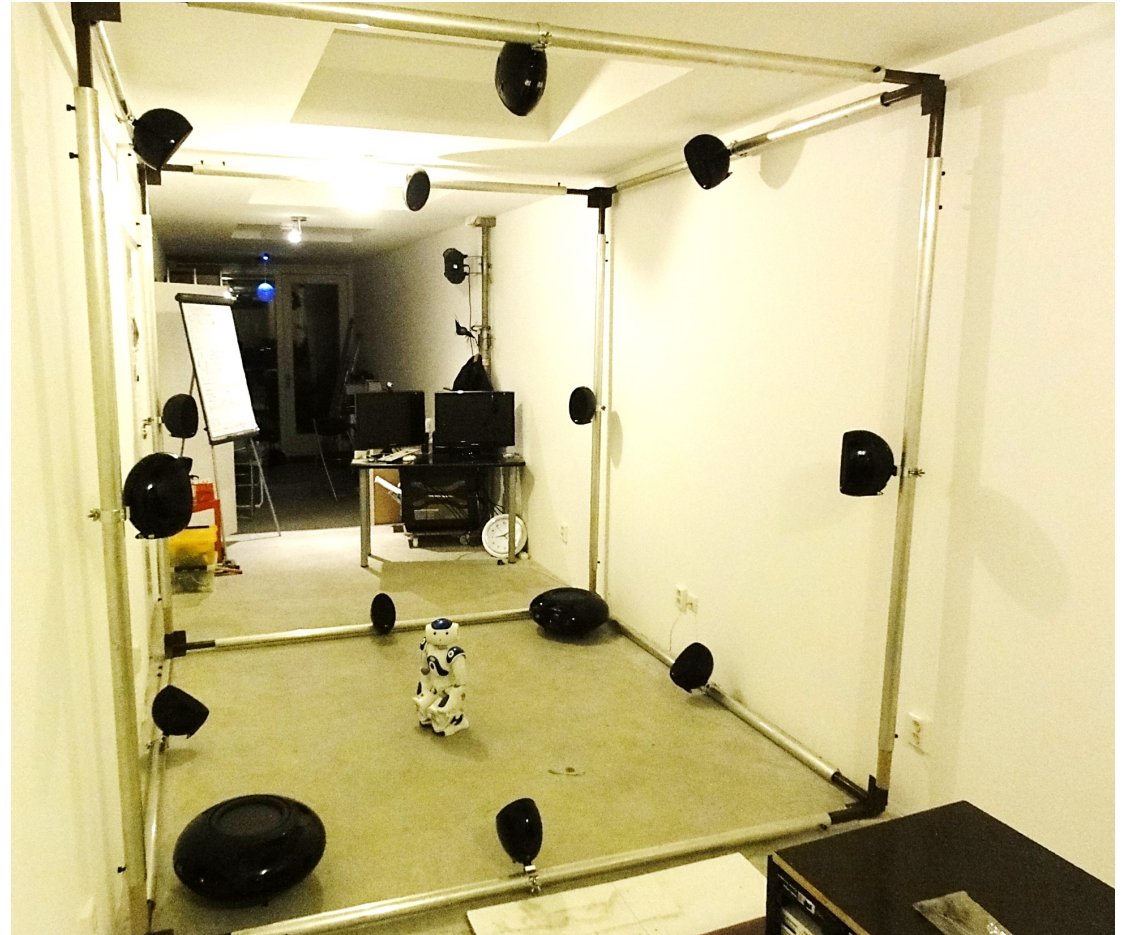
☐ Spatial audio

Some issues:

- Acoustic attenuation
- Occlusion
- Reflection



Ambisonics in my home lab: fourteen surround-sound speakers in a 3D cuboctahedron geometry (and a NAO robot listening)

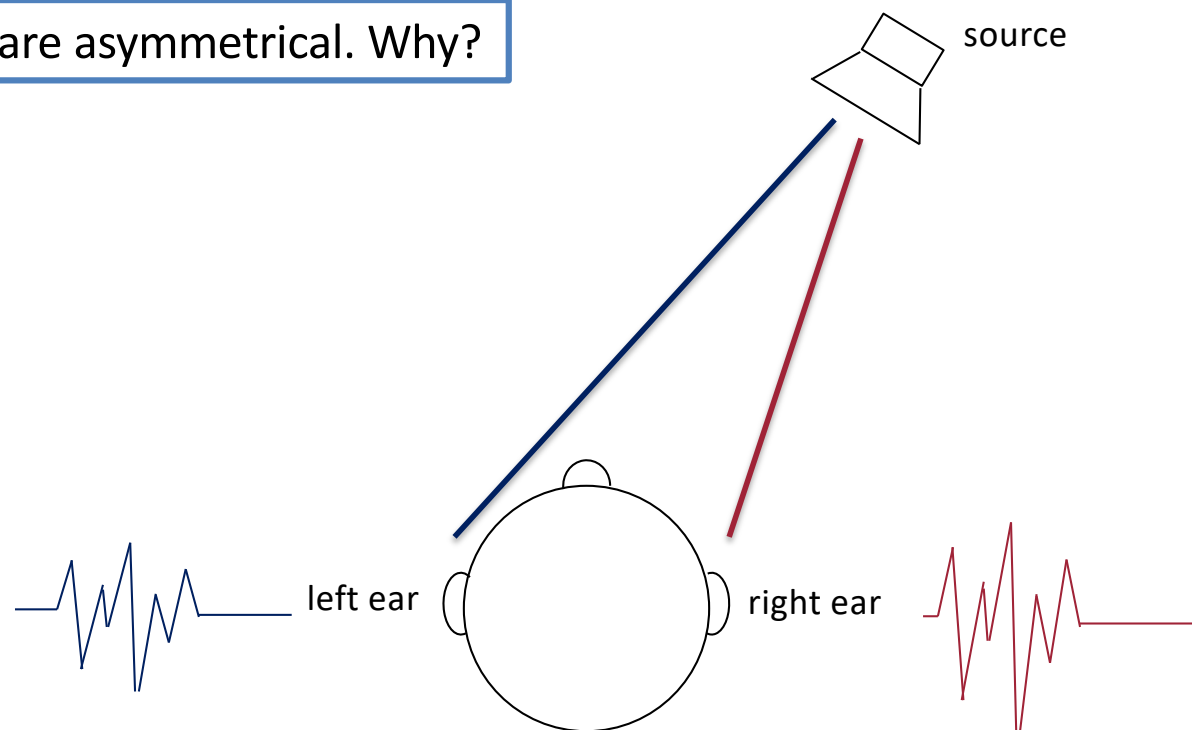


Output devices

❑ Spatial audio

- Acoustic attenuation: spatial perception of the acoustic source

Owl ears are asymmetrical. Why?



Output devices

❑ Spatial audio

- Acoustic attenuation: spatial perception of the acoustic source



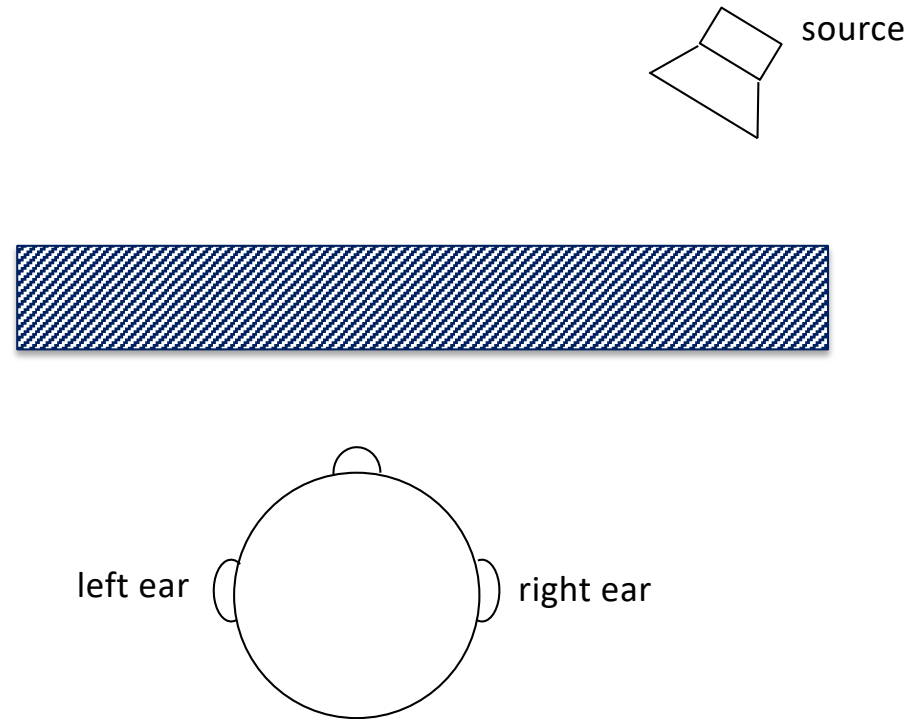
Owl ears are asymmetrical for greater precision: horizontally $\leftrightarrow$ and vertically $\updownarrow$



Output devices

□ Spatial audio

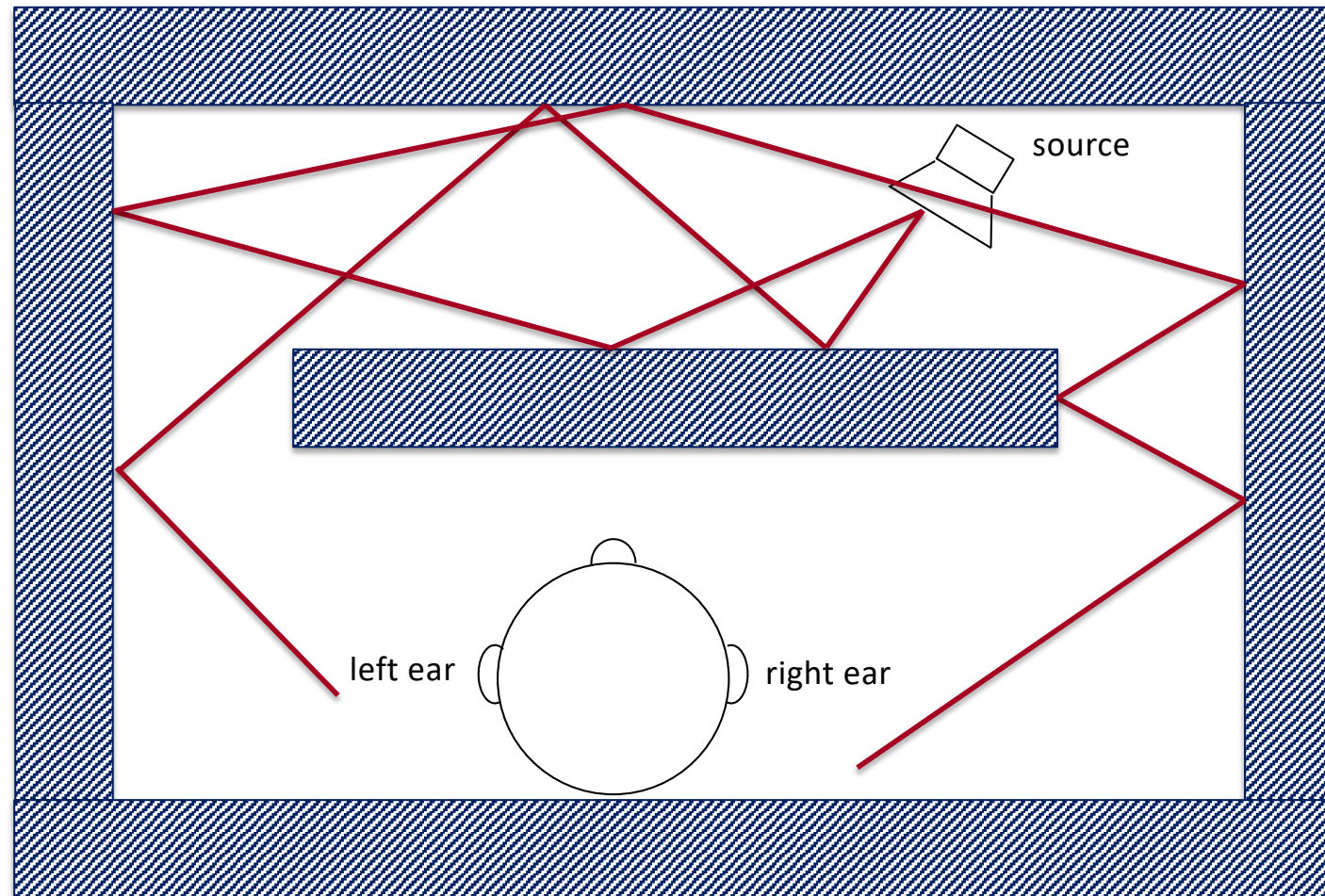
- Occlusion



Output devices

❑ Spatial audio

- Reflection / echo



Output devices

❑ Spatial audio: for your perusal

NTRS

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3-D Sound for Virtual Reality and Multimedia

NTRS Full-Text: [Click to View](#) [PDF Size: 12.9 MB]

Author and Affiliation: Begault, Durand R. (NASA Ames Research Center, Moffett Field, CA United States)
Trejo, Leonard J. [Technical Monitor]

Abstract: Technology and applications for the rendering of virtual acoustic spaces are reviewed. Chapter 1 deals with acoustics and psychoacoustics. Chapters 2 and 3 cover cues to spatial hearing and review psychoacoustic literature. Chapter 4 covers signal processing and systems overviews of 3-D sound systems. Chapter 5 covers applications to computer workstations, communication systems, aeronautics and space, and sonic arts. Chapter 6 lists resources. This TM is a reprint of the 1994 book from Academic Press.

Publication Date: Aug 01, 2000

Document ID: 20010044352 (Acquired Jun 02, 2001)

Subject Category: MAN/SYSTEM TECHNOLOGY AND LIFE SUPPORT

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☐ Begault, Durand R.

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Publication Year

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Document Type

☐ Technical Report

NASA Center

☐ Ames Research Center

Organization

☐ NASA Ames Research Center

Subject Category

☐ MAN/SYSTEM TECHNOLOGY AND LIFE SUPPORT

Subject Terms

☐ ACOUSTICS

☐ HEARING

☐ HUMAN-COMPUTER INTERFACE

☐ INTERPROCESSOR COMMUNICATION

☐ MULTIMEDIA

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☐ THREE DIMENSIONAL MODELS

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Advanced I/O devices



Robert Downey Jr. volunteers to be voice of Mark Zuckerberg's Jarvis
(The Mercury News)



Ironman JARVIS
S.H.I.L.D. OS
Created By Yash1331

Chrome

Control Panel

VLC Player

Mozilla Firefox

uTorrent

Skype

RAM USAGE

50%

PROC: 0

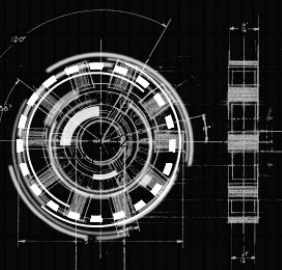
Used: 2.0 GB

Free: 2.0 GB

RAM

653.5 GB - 50.0 B/s 149.2 GB - 0.0 B/s

STARK INDUSTRIES



SYSTEM

CPU Usage 32%

RAM Usage 50%

SWAP Usage 53%

HD C: 31% 89% 16%

HD D: 31% 89% 16%

HD E: 31% 89% 16%

HD F: 31% 89% 16%

HD G: 31% 89% 16%

HD H: 31% 89% 16%

HD I: 31% 89% 16%

HD J: 31% 89% 16%

HD K: 31% 89% 16%

HD L: 31% 89% 16%

HD M: 31% 89% 16%

HD N: 31% 89% 16%

HD O: 31% 89% 16%

HD P: 31% 89% 16%

HD Q: 31% 89% 16%

HD R: 31% 89% 16%

HD S: 31% 89% 16%

HD T: 31% 89% 16%

HD U: 31% 89% 16%

HD V: 31% 89% 16%

HD W: 31% 89% 16%

HD X: 31% 89% 16%

HD Y: 31% 89% 16%

HD Z: 31% 89% 16%

HD AA: 31% 89% 16%

HD AB: 31% 89% 16%

HD AC: 31% 89% 16%

HD AD: 31% 89% 16%

HD AE: 31% 89% 16%

HD AF: 31% 89% 16%

HD AG: 31% 89% 16%

HD AH: 31% 89% 16%

HD AI: 31% 89% 16%

HD AJ: 31% 89% 16%

HD AK: 31% 89% 16%

HD AL: 31% 89% 16%

HD AM: 31% 89% 16%

HD AN: 31% 89% 16%

HD AO: 31% 89% 16%

HD AP: 31% 89% 16%

HD AQ: 31% 89% 16%

HD AR: 31% 89% 16%

HD AS: 31% 89% 16%

HD AT: 31% 89% 16%

HD AU: 31% 89% 16%

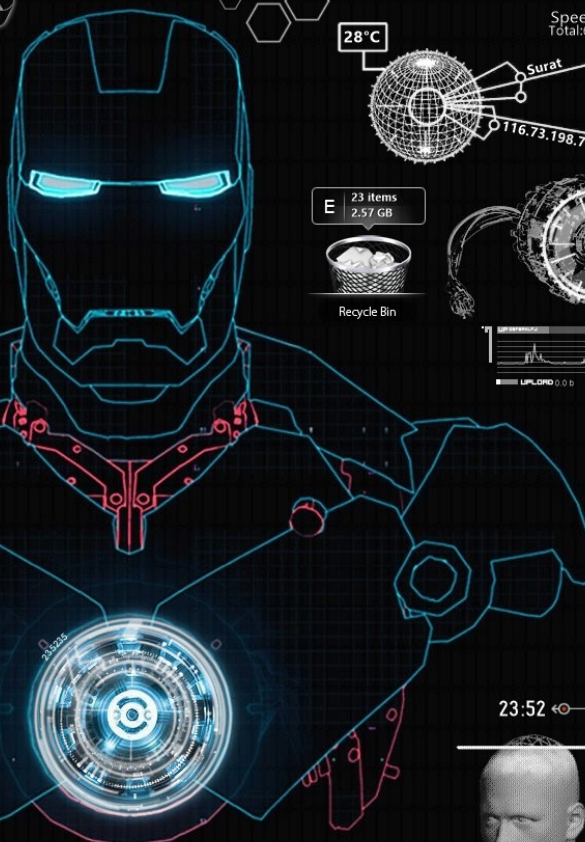
HD AV: 31% 89% 16%

HD AW: 31% 89% 16%

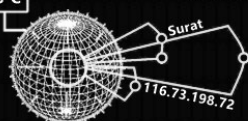
HD AX: 31% 89% 16%



GOOGLE
GMAIL
FACEBOOK
YOUTUBE
IMDB
KAT
WEB
9GAG
PIRATE BAY
STUMBLE UPON
YAHOO
PUPIL POD
WIKIPEDIA



28°C



Speed: 0.0 B

Total: 653.5 GB

DWN

23:52

Tuesday

August 20, 2013

FILESYSTEMS:

HDD C: 280.7 GB

HDD D: 13.2 GB

HDD F: 0.0 GB

HDD E: 0.0 GB

HDD H: 0.0 GB

DISK

C:\ 280.7 GB/916.7 GB used

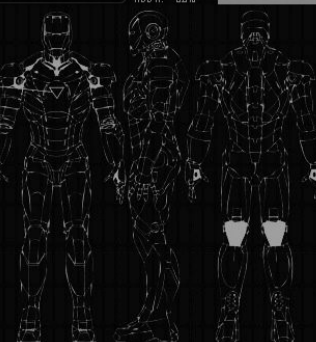
D:\ 13.2 GB/14.7 GB used

23 items

2.57 GB

E

Recycle Bin

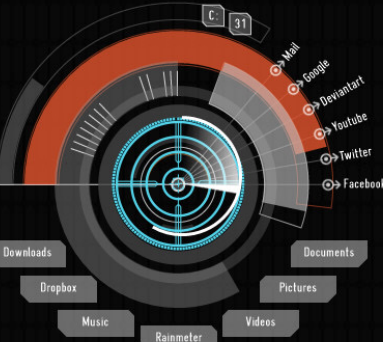


100
80
60
40
20
20
40
60
80
100
100
80
60
40
20
20
40
60
80
100



Currently power level is at 100 percent and holding steady.

23.52



Su Mo Tu We Th Fr Sa

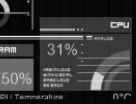
18 19 20 21 22 23 24

BATTERY

AC LINE 100%

no battery

20-Aug., Tuesday
11:52 PM



TIME 10:10

11:52

20.08.13

HOURS

35

35

35

35

35

35

35

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35

35

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35

35

35

35

0:00

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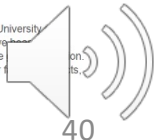
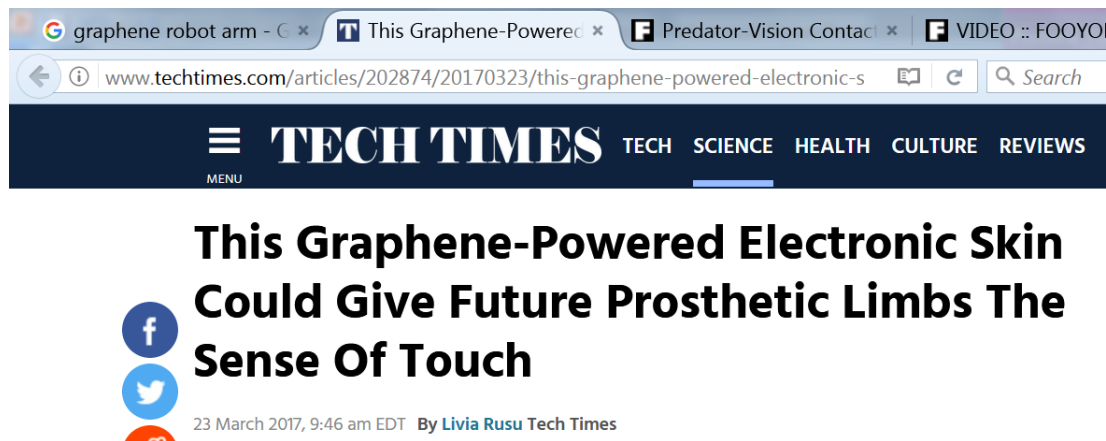


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Advanced I/O devices

❑ Advanced and futuristic

- The trend
 - Blending input and output
(e.g., tablets and phones, VR headsets, wearable gadgets, etc.)
 - More intuitive and natural interactions
(e.g., NLP, touch interface, motion-tracked controllers, etc.)
 - Multimodal interaction
 - More personalized experience (potentially dangerous?)
- Robotics, BCI, Human Augmentation



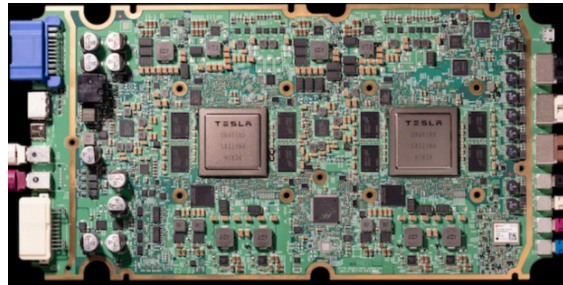
Advanced I/O devices

❑ Advanced and futuristic

- Blending input and output
 - It becomes harder to find a pure input/output device
 - Every gadget is a “multimedia PC” since the 90s



Apple: With a powerful Apple-designed H1 chip in each cup, our custom acoustic design, and advanced software, AirPods Max use computational audio to create a breakthrough listening experience.



Source: Oculus



Source: Apple

Advanced I/O devices

❑ Advanced and futuristic

- Intuitive and natural interaction

- NYT - Is it too soon to give my kid a tablet?

- <https://www.nytimes.com/2020/04/17/parenting/tablet-child-screentime.html>

- The Guardian - Are iPads and tablets bad for young children?

- <https://www.theguardian.com/society/2014/jan/08/are-tablet-computers-bad-young-children>

- Multimodal interaction

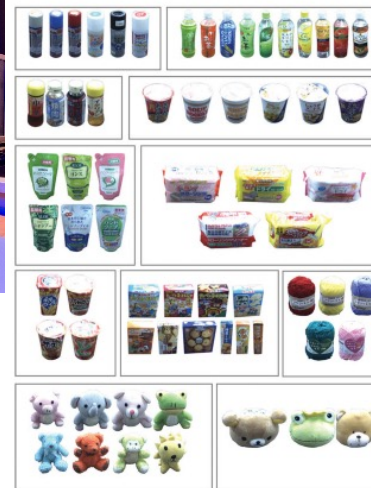


Virtuix Omni One VR treadmill



Multimodal interface for children with severe intellectual disability

A robot's multimodal learning (Tangiuchi et al. 2019)



(a)



(b)



(c)



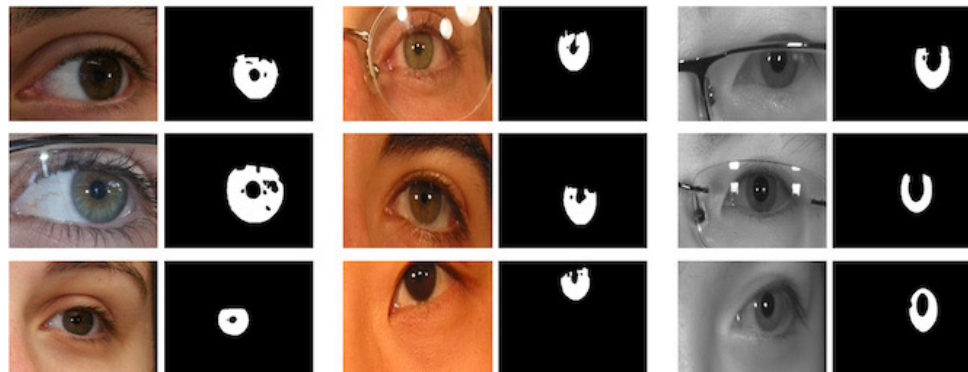
(d)

Advanced I/O devices

❑ Personalized experiences

- Handwriting recognition/personal digital assistants
 - 3M Palm Pilot, Go Corp., Sony, Toshiba
- Smart card
 - thin plastic card, embedded μ processor and memory
 - information about a user (e.g., employee ID, credit details, etc.) is stored on the card
 - outputs information to special card readers
- Biometric devices
 - advanced smart card that contains characteristics about a user such as fingerprints, voice prints, retina prints, or signature dynamics

At-A-Distance Iris Segmentation and Recognition



UBIRIS.v2

FRGC

CASIA.v4-distance

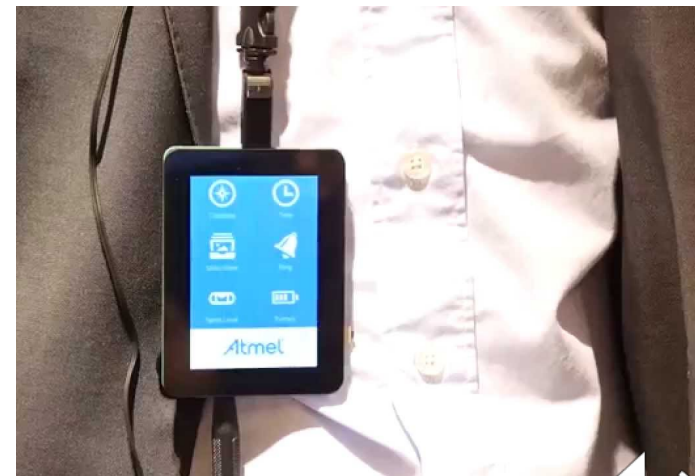
Prof. Ajay Kumar



Advanced I/O devices

❑ Personalized experiences

- Texture sensation
 - sandpaper system (MIT)
 - uses a motor-driven, force-feedback joystick that uses tiny virtual springs to simulate motion while the user moves the joystick over patches of computationally created textures displayed on a screen
- Gesture, speech, and gazing
 - two-handed gesturing by voice, and selection by gaze (CMU)
 - “Turn that block upside down.”
- Person tracking
 - Active Badge™ system (Olivetti/DEC)
 - tracks people inside a building
 - used as a communications device
 - can be turned off for privacy



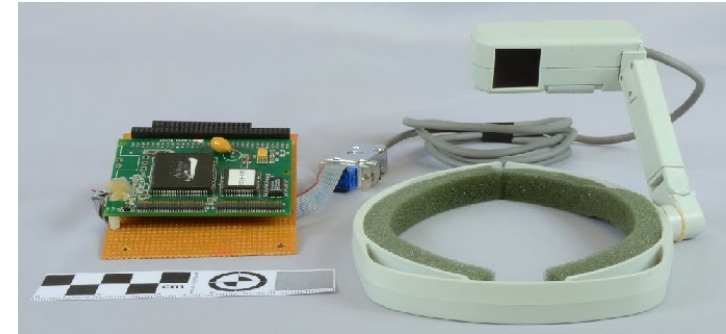
Atmel Smart Badge Demonstrator



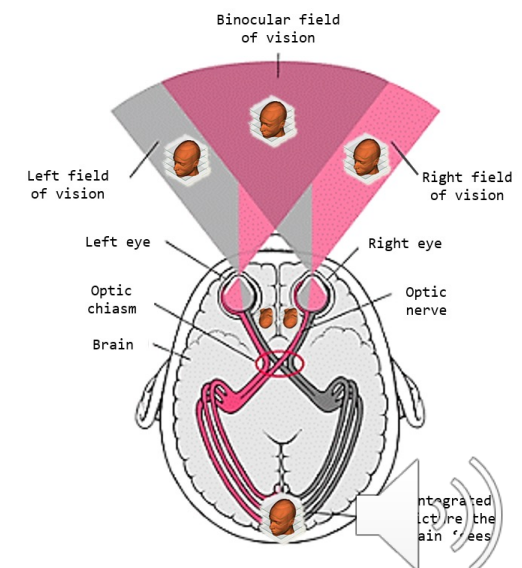
Advanced I/O devices

❑ Wearables

- Wearable computer
 - Private Eye™ (Reflection Technology)
 - user wears a single high-resolution LCD over one eye, while looking out the other eye; image projected at infinity
 - coupled with a portable computer, and other input devices
 - Wearable Computer™ (Computing Devices International)
 - portable, body-mounted, voice-activated computer
 - tested in Bosnia, presently being adopted by the U.S. Armed Services
 - Sixth Sense
 - combination of camera and projector
- Helmet-mounted display with speech interface
 - military applications, “aim-fire” scenarios



↑
Display covers one eye,
yet user has illusion that
display is transparent.
How come? (Lecture 03)



Advanced I/O devices

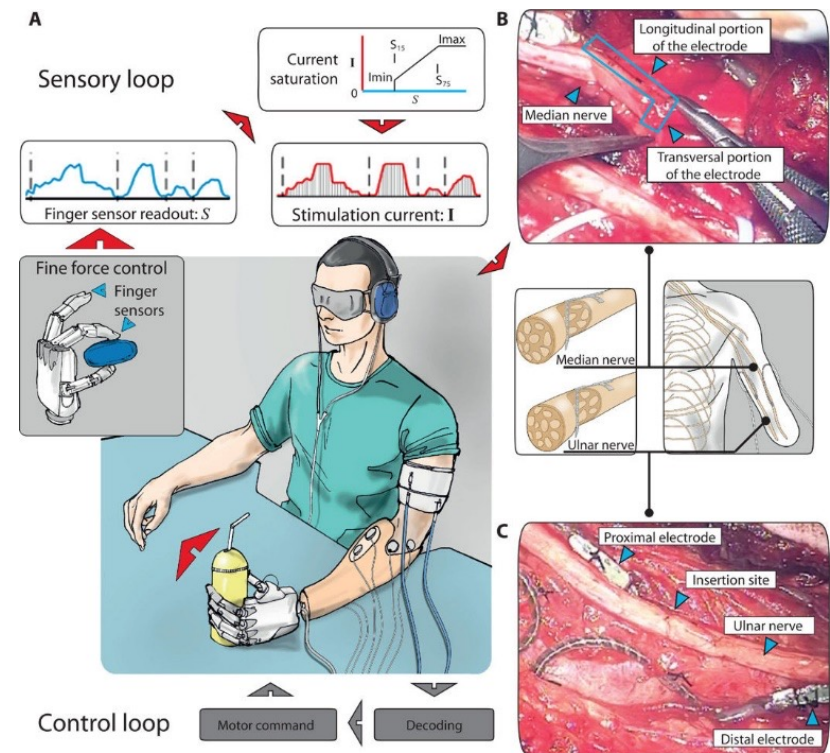
□ Robotics & Human Augmentation

- Compensate for physical disability
- Tissue-synthetic interfaces connecting the biological body to a machine

□ BCI & Human Augmentation

- Compensate for neurological conditions (e.g., dementia, Parkinson's disease, etc.)
- BCI connecting brain to a machine (and machine is connected to brain)

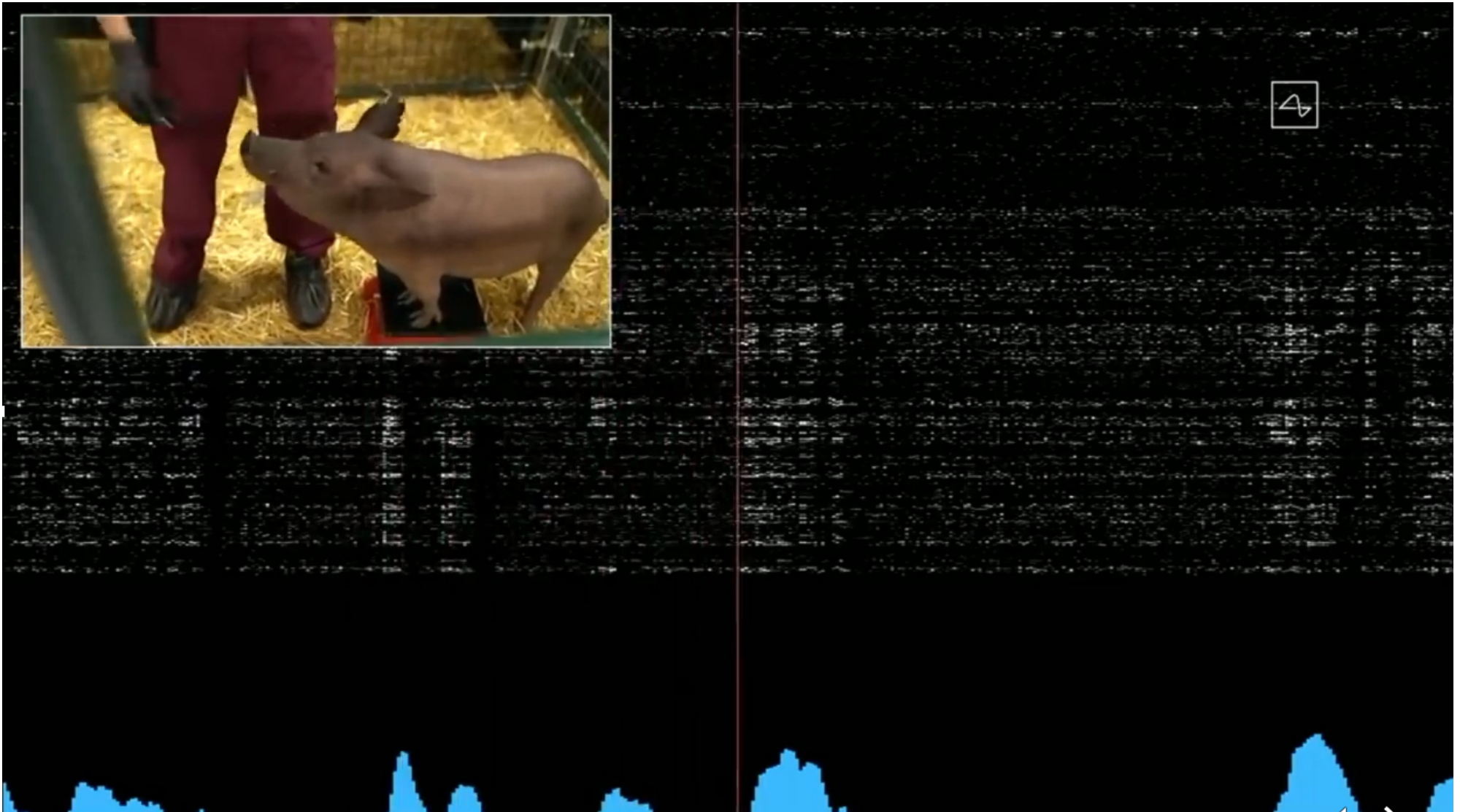
<https://www.youtube.com/watch?v=PB1gDYkvU3w>



Source: sciencemag.org



Brain-Computer Interfaces: neural implants

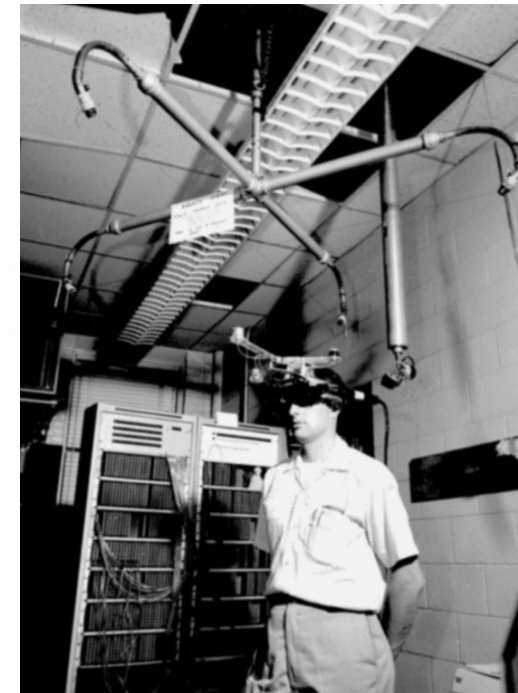
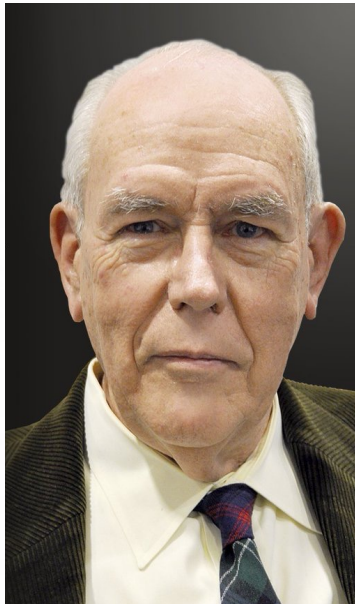


Brain signal visualized and sonified (cf. 'problem representations')



Advanced I/O devices

- ❑ Full body involvement



Remember Ivan Sutherland

Kat VR Announces New 'Mini' Version of Its VR Treadmill, Kickstarter Coming Soon

By Scott Hayden - Mar 23, 2018 15

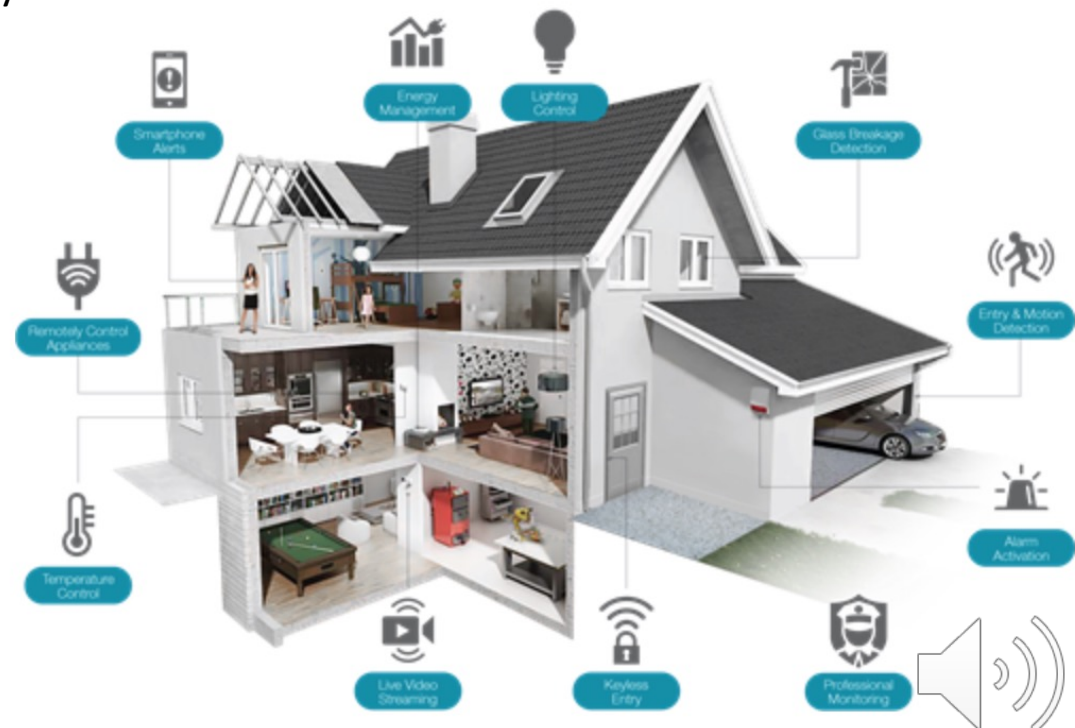
image courtesy Kat VR



Advanced I/O devices

☐ Smart living

- Smart rooms
 - can identify people and interpret their actions
 - house that knows where your kids are and tells you if they are getting into trouble
 - can supervise students during exams ;-)
 - research being conducted globally
 - Person Finder - Pfinder
 - video cameras to recognize faces, expressions, gestures
 - mics for speech recognition
- Smart home
 - performs activities according to user's preferences and usual actions



Advanced I/O devices

❑ Smart textiles

- Smart clothes
 - sort of a personal assistant that you wear
 - tells you the name of people you meet, directions to your next meeting, etc.
 - built-in computer, camera, microphones, other sensors
 - camera built into the frame of eyeglasses that captures images
 - face-recognition software that tells you the name of the person you are looking at by whispering his or her name into your ear



We Came In Color LED T-Shirt

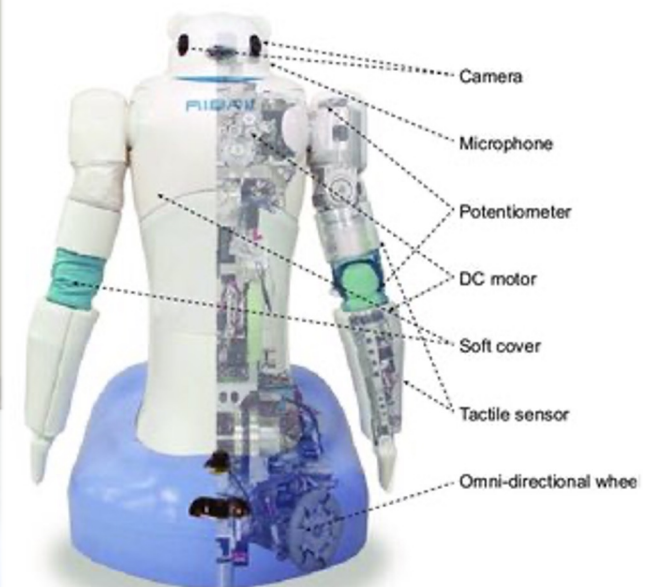


Advanced I/O devices

❑ Social robots



RIBA (Robot for Interactive Body Assistance) lifts up to 61kg/134lbs



Ergonomic aspects? For whom? Who may like it? Who not?



Future I/O devices

❑ Virtual Reality

Comment on Chief John Anderton's interface from an ergonomic viewpoint

May cause injuries:

- 1 wave hands and arms around
- 2 at shoulder level
- 3 for extended periods of time

Think of an application where this would make perfect sense ergonomically:
If handling the interface is meant as a workout in an e-health environment

Spielberg, S. (2002). *Minority Report*. Los Angeles, CA: 20th Century Fox, DreamWorks Pictures.



Welcome to the future: Bionic Brains

- Human-to-human communication?
- Brain-to-brain communication!



0, 1, 1, 0, 0, ...
Internet

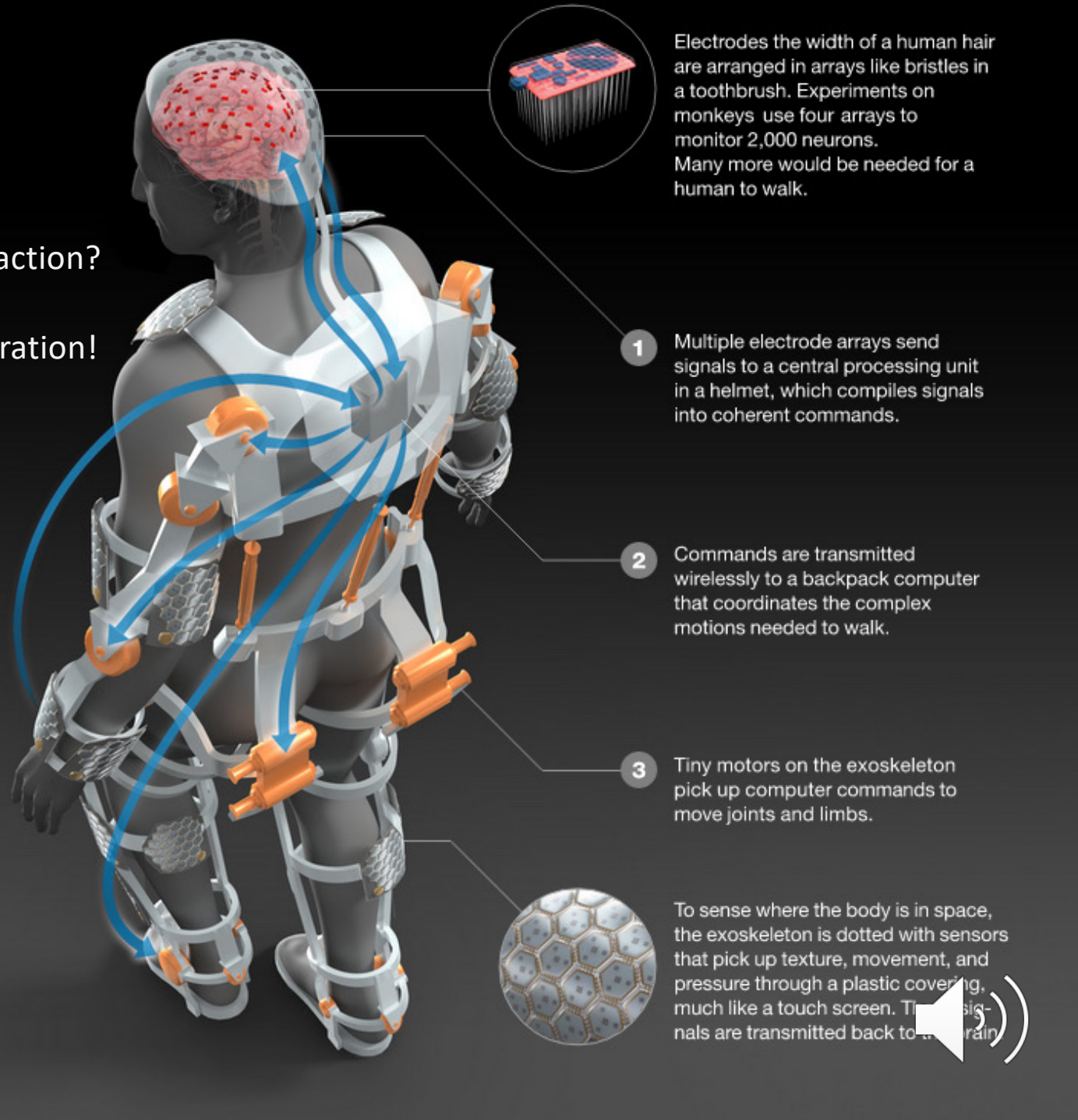


Grau, C., Ginhoux, R., Riera, A., Nguyen, T. L., Chauvat, H., Berg, M., ... & Rutlini, G. (2014). Conscious brain-to-brain communication in humans using non-invasive technologies. *PloS one*, 9(8), e105225.

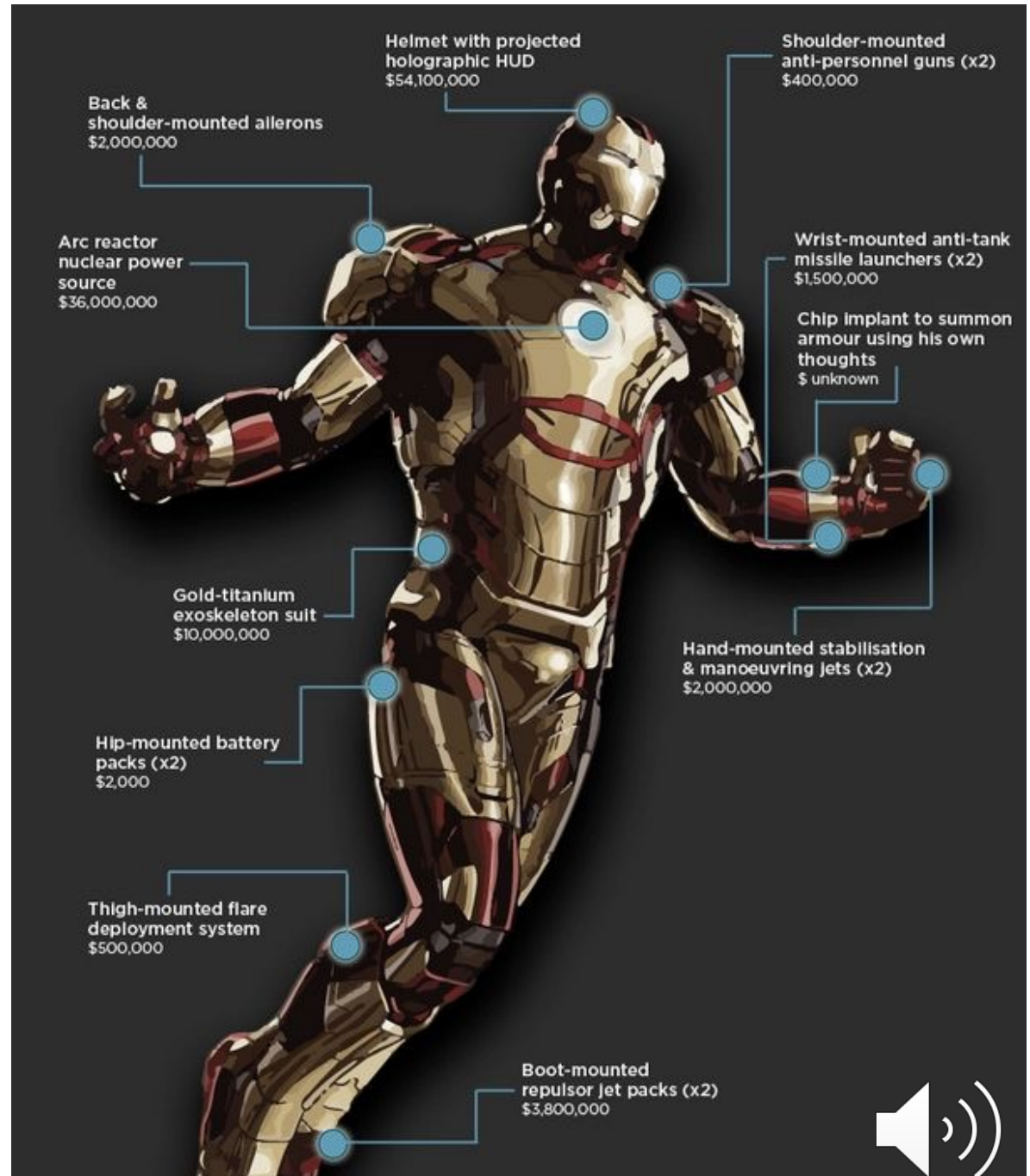


Bionic Brains

- Human-machine interaction?
- Human-machine integration!



Iron Bodies?



However...

Transcranial Direct Current Stimulation

Reviews and perspectives

Electrified minds: Transcranial direct current stimulation (tDCS) and Galvanic Vestibular Stimulation (GVS) as methods of non-invasive brain stimulation in neuropsychology—A review of current data and future implications

Kathrin S. Utz^{a,c,*}, Violeta Dimova^{a,c}, Karin Oppenländer^b, Georg Kerkhoff^{a,c,*}

Shaping Memory Accuracy by Left Prefrontal Transcranial Direct Current Stimulation

Bastian Zwissler,¹ Christoph Sperber,¹ Sina Algedinger,¹ Sebastian Schindler,² Johanna Kissler,² and Christian Plewnia^{1,3}

Review

Treatment of depression with transcranial direct current stimulation (tDCS): A Review

Michael A. Nitsche^{a,*}, Paulo S. Boggio^b, Felipe Fregni^c, Alvaro Pascual-Leone^{a,d}

Reviews and perspectives

Evidence that transcranial direct current stimulation (tDCS) generates little-to-no reliable neurophysiologic effect beyond MEP amplitude modulation in healthy human subjects: A systematic review

Jared Cooney-Henry^{a,*}, Jason D. Fortin^a, Olivia Carter

University of Melbourne, School of Psychological Sciences, Melbourne, VIC, Australia

Research report

Transcranial direct current stimulation of frontal cortex decreases performance on the WAIS-IV intelligence test

Kristin K. Sellers^{a,b}, Juliann M. Mellin^a, Caroline M. Lustenberger^a, Michael R. Boyle^{a,c,d}, Won Hee Lee^d, Angel V. Peterchev^{a,b,e}, Flavio Frohlich^{a,b,c,d,h,*}



loc.us tDCS v2



Ghostbusters (1984)



Bionic Bodies

Dictionary

Search for a word

biochip
/'bɪəʊtʃɪp/

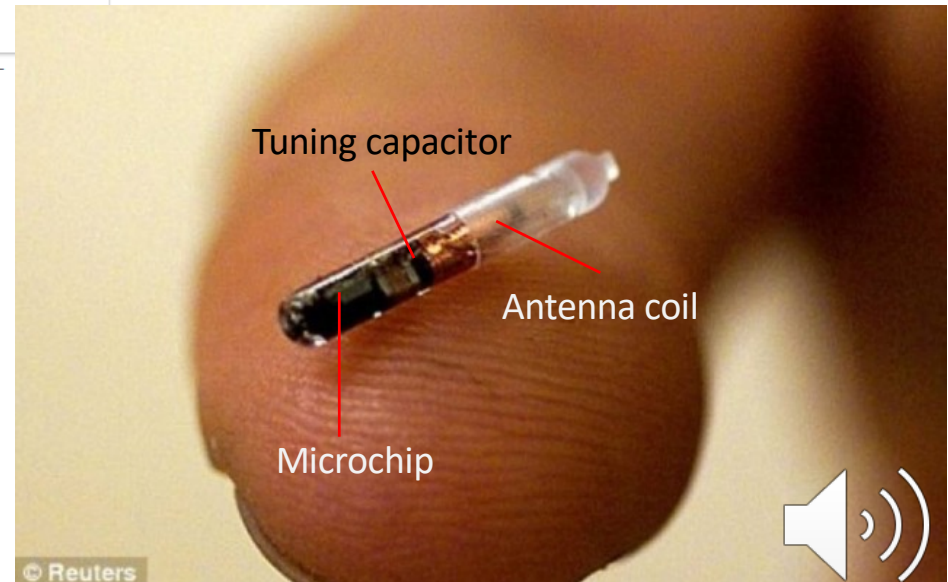
noun
plural noun: **biochips**

a microchip designed or intended to function in a biological environment, especially inside a living organism.

- a logical device analogous to the silicon chip, whose components are formed from biological molecules or structures.

Translations, word origin, and more definitions

- ☐ Biochip reader sends radio signals,
- ☐ causing low-power electromagnetic field,
- ☐ activating the biochip,
- ☐ which sends back its ID through radio
- ☐ A/D conversion in reader
- ☐ Reader outputs response (“Hello, Jailu”) (e.g., speech, actuator, screen)



Bionic Bodies

nature
International journal of science

Review Article | Published: 14 December 2016

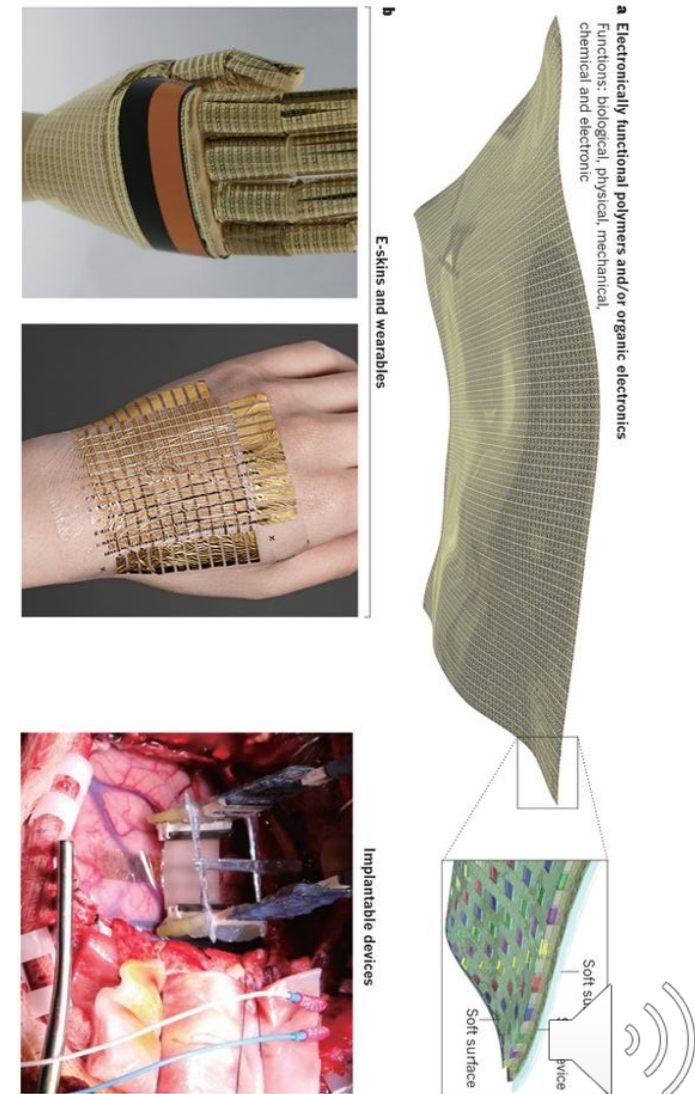
The rise of plastic bioelectronics

Takao Someya , Zhenan Bao & George G. Malliaras

Nature **540**, 379–385 (15 December 2016) | [Download Citation](#) 

Abstract

Plastic bioelectronics is a research field that takes advantage of the inherent properties of polymers and soft organic electronics for applications at the interface of biology and electronics. The resulting electronic materials and devices are soft, stretchable and mechanically conformable, which are important qualities for interacting with biological systems in both wearable and implantable devices. Work is currently aimed at improving these devices with a view to making the electronic–biological interface as seamless as possible.



Biohybrid Circuits



Concept | [Full Access](#)

Electrical Interfacing of Nerve Cells and Semiconductor Chips

Peter Fromherz Prof. Dr. [✉](#)

First published: 15 March 2002

| [https://doi.org/10.1002/1439-7641\(20020315\)3:3<276::AID-CPHC276>3.0.CO;2-A](https://doi.org/10.1002/1439-7641(20020315)3:3<276::AID-CPHC276>3.0.CO;2-A) | Cited by: 87

[PolyU eLink](#)

Peter Fromherz is a member of the Editorial Advisory Board for ChemPhysChem.

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Abstract

The electrical interfacing of individual nerve cells and silicon microstructures is considered, as well as the assembly of elementary hybrid systems made of neuronal networks and semiconductor microelectronics. Without electrochemical processes, coupling of the electron-conducting semiconductor and the ion-conducting neurons relies on a close contact of cell membrane and oxidised silicon with a high resistance of the junction and a high conductance of the attached membrane. Neuronal excitation can be elicited and recorded from the chip by capacitive contacts and by field-effect transistors with an open gate. Integrated ionic-electronic systems are obtained by the outgrowth of neuronal networks on the surface of the silicon chip, by implementing electrical circuits in the chip and by two-way interfacing of the neuronal and the electronic components.

Both brains and computers work electrically. It is an intellectual and technological challenge to join the two systems directly as sketched in Figure 1.1. When we study that problem today, we are not concerned whether it can be really solved in the foreseeable future in the sense that brain tissue is integrated in computers or that semiconductor chips are integrated in

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VIDEO

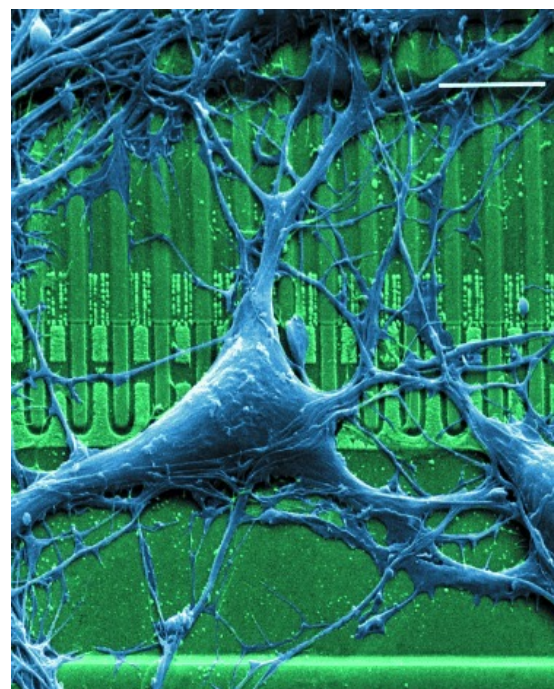
ROLL THE DICE

SUBSCRIBE

SCIENCE

Silicon Chips Wired With Nerve Cells Could Enable New Brain/Machine Interfaces

By Clay Dillow March 22, 2011



Nerve cell from rat brain (hippocampus) on a silicon chip. Colored electron micrograph (scale bar 10 μm). The surface of the chip is made of silicon dioxide. The metal-free gates of a linear array of field-effect transistors are visible as dark squares. The neuron was cultured for several days in an electrolyte.



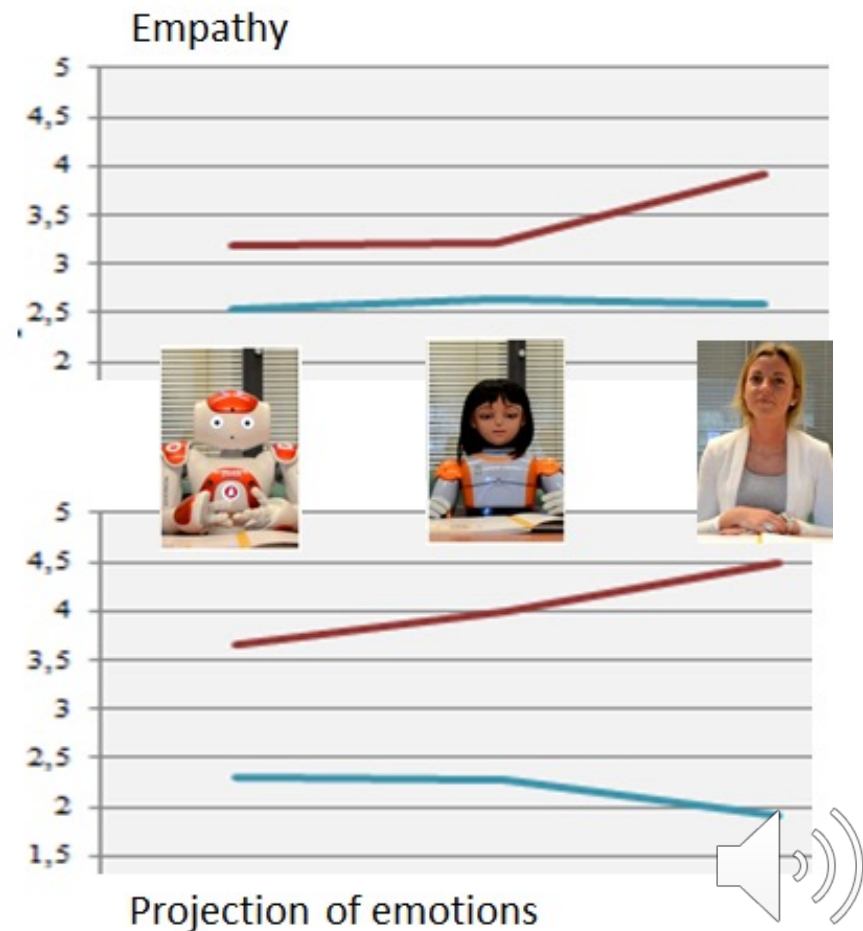
Agency-centered Design

- Bionic people
- Machines with biochips
- The distinction between virtual and real humans becomes obsolete. The transition is gradual
- 'Me' as 'identity of self' is not restricted to the organic body anymore. Parts can be exchanged or upgraded by technology
- Psychologically, people attribute agency (including emotion) to non-living artifacts. Deliverance of a function is important, whether done by a human being or a machine is not



Agency-centered Design

- People attribute agency (including emotion) to non-living artifacts



Agency-centered

- People attribute agency (incl living artifacts



International Journal of Psychology

OR1528

Empathy with and Projecting Feelings onto Robots from Schemas about Humans

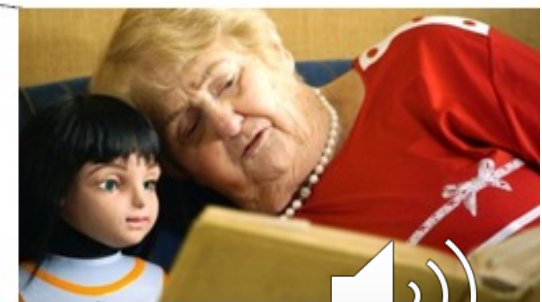
Elly A Konijn, Johan F Hoorn Vrije Universiteit Amsterdam, Netherlands

Quintessential to the rise of social robots in service-oriented professions such as health-care, hospitality, and education is the emotional relationship a user establishes with the machine. For example, detailed facial expressions may evoke empathy and emotions, similar to what we feel for humans. But what do we feel when robots are mistreated? We conducted a 3 (robot/human: Robokind's "Alice" vs. Alebaran's Nao vs. human) x 2 (treatment: positive vs. negative) between-subjects experiment. Consequently, six video clips showed either a robot or a human being treated kindly or aggressively by a human actor. Randomly assigned participants (N=265) were measured for empathy, emotional responsiveness, and emotion attribution. Participants indeed showed empathy and emotions toward robots and endowed them with emotions, yet a little less than they did with humans. Results also indicated slightly stronger responsiveness toward more expressive faces (i.e. Alice). Implications for future research and possibilities for service professions are discussed.



Agency-centered Design

- Deliverance of a function is important, whether done by a human being or a machine is not



*The In-between Machine
The Unique Value Proposition of a Robot or Why we are Modelling the Wrong
Things*

Johan F. Hoorn¹, Elly A. Konijn^{1,2}, Desmond M. Germans³, Sander Burger⁴ and Annemiek Munneke⁴
¹Center for Advanced Media Research Amsterdam, VU University, De Boelelaan 1081, Amsterdam, Netherlands
²Dept. of Communication Science, Media Psychology, VU University Amsterdam, Netherlands
³Germans Media Technology & Services, Amsterdam, Netherlands
⁴KeyDocs, Amsterdam, Netherlands

Agency-centered Design

- Social robots will become the universal interface between humans and data analytics in the IoT

Neuromorphic Computers

Brain implants

IoT with biochips

"Granny, I'll be your universal interface to the overload of data of your devices, apps, and Internet sources"

"The zip code is 1057 BA"
"AccuWeather says it is nice outside"
"Can you open the window? Stale air here"
"This produce has sugar in it"
"My GPS tells me, we should go right"
"Smart Thermostat says you should turn down the heat a little"
"Time to take your Prednisone"
"Shall we play chess?"

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International Robotics & Automation Journal

The ultimate android

For to us an Android is born...
Imagine a machine
Human nor humane
Acting on your behalf
imagining you as human -
a machine capable of being

Technical Paper

Volume 4 Issue 1 - 2018

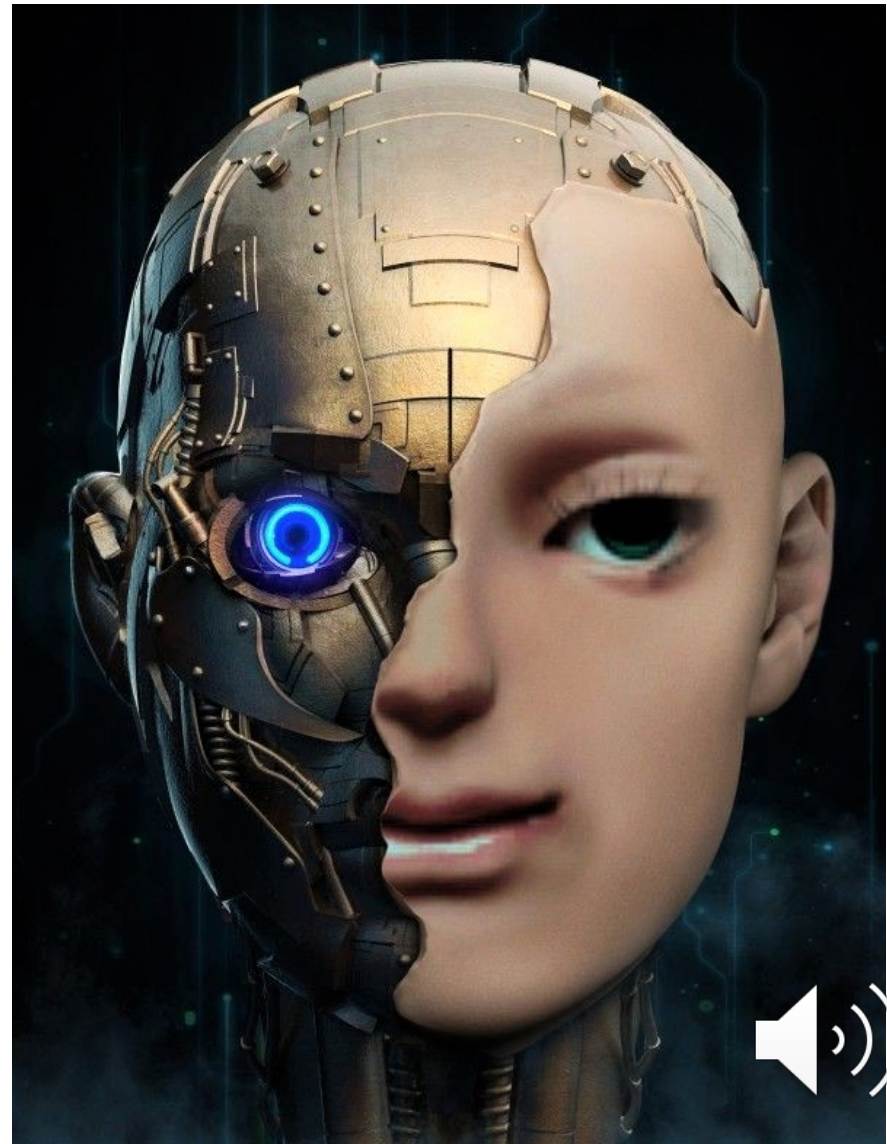
Johan F Hoorn^{1,2}
¹Department of Computing and School of Design, The Hong Kong Polytechnic University, Hong Kong
²Department of Communication Science, Vrije Universiteit

69
HEAT SET TO
08

Conclusion

We are heading for:

- Brain-to-brain communication
- Human-machine integration
- Virtual Reality:
 - Neuromorphic biochip input
 - Induction of (false) memories
 - Simulated partners
- Agency-centered design
- A new kind of psychology!



Human-generated summary



HCI largely developed from Ergonomics (Human Factors) because it is concerned with human performance and well-being in working with (computing) machinery

I/O Devices, from a simple mouse to advanced VR goggles and gloves, directly concern Ergonomics because they are the interface to the computational system. If they don't work, your system runs down

Ergonomics is mainly known for its physical aspects, e.g., preventing mouse arms. A disaster coming from a lack of physical ergonomics was the 'big finger' incident at the Tokyo stock exchange (Lecture 01)

Human-generated summary



Cognitive ergonomics focuses on e.g., memory load (data overflow), attention, ambiguity, or the effects of warnings. A disaster coming from a lack of cognitive ergonomics was the crash of the American Airlines Flight 965 (Lecture 01)

I/O Devices may tend to all sensory channels humans have (visual, acoustic, haptic, even smell). For a device to be ergonomically functional, you have to study the human physiological, perceptual, and information processing systems (Lecture 02 and 03). Otherwise, you don't know how it works and make design errors

While choosing I/O Devices, be aware of the following (next slide):



Human-generated summary



A device should match physiological / psychological characteristics of users, their training, and their expertise. E.g., older adults may be hampered with arthritis, and may be unable to type; inexperienced users may be unfamiliar with BCIs and VR treadmills, etc.

Devices should be appropriate for the tasks to be performed. E.g., a drawing task requires a device that allows continuous movement; selecting an option from a list requires a device that permits discrete movement, etc.

Devices should be suitable for the intended work and environment. E.g., speech input may be useful when there is no keypad or keyboard, or when the eyes and hands are busy, but is unsuitable in noisy conditions such as a business fair



Exam Sample Item

Category “Knowledge”

Which of the following is an example of an ergonomic risk factor?

A Neutral postures

B Rest



C Repetition

D Personal protective equipment



- Spread repetitive work out during the day
- Take stretch pauses
- Rotate task with co-workers if possible
- Change hands or motions frequently



References

Lederman, S. J., & Klatzky, R. L. (2009). Haptic perception: A tutorial. *Attention, Perception, & Psychophysics*, 71(7), 1439-1459.

<https://link.springer.com/content/pdf/10.3758%2FAPP.71.7.1439.pdf>

Every Mac Startup & Crash Chime: <https://youtu.be/n23dp8caq9A>



Let's exercise! Shoulder stretch

